

PROVISIONAL PATENT APPLICATION

Lift Platforms, Transport Vehicles, Hoisting and Elevator Systems, Launch and Station-Keeping Systems, Orbital Momentum-Management Systems, Marine Trim Systems, and Soft-Landing Systems Built from Calibrated Weight-Trim Cells, and Methods of Operating Them

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This specification is complete and self-contained. Applicant's companion applications are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no lift, ascent, station-keeping, orbit change, ballast replacement, or other performance is represented as achieved, and no response sign, magnitude, additivity, retained state, or scaling law is presumed. Production-map embodiments operate only at operating points their stored, independently validated response maps cover; the separately claimed POAMS-native direct-operation species does not require a preexisting nonzero response map.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media. A companion draft ("Support-Force Control Apparatus Using a Resonantly Driven Ordered Medium, Calibrated Support-Force Modules and Plural-Module Support-Force Systems, and Methods of Operating, Qualifying, and Recalibrating Them") discloses the single-cell apparatus and its factory qualification in greater depth; other companions address measurement instruments, energy infrastructure, media and array manufacturing, information devices, time-rate metrology, and completion-scheduled computation. Each is nonessential background. The weight-trim cell, its stored response map, its qualification, and its safe-off behavior are described in this specification to the extent the systems and methods claimed here require.

FIELD

Multimodal vehicles, lift and propulsion systems, hoisting machinery, launch systems, spacecraft momentum management, marine trim systems, and descent systems; more particularly, systems in which the support force (weight) of one or more driven bodies of ordered medium, and thereby the net support force of a frame, vehicle, load, hull, or descending body that rigidly bears them, is commanded from a

stored, empirically established response map by calibrated **weight-trim cells**, replacing or augmenting wheels, rotors, tracks, wings, ballast, counterweights, propellants, and momentum-storage hardware; and methods of operating such systems.

BACKGROUND

Terrestrial transport rests on the wheel and its friction pair; vertical transport on counterweights and winches; marine transport on buoyancy, ballast, and screw propulsion; flight on momentum exchange with air; spaceflight on propellant expulsion and reaction wheels. Each is bounded by its medium: rolling resistance and grade, counterweight mass, draft and ballast capacity, aerodynamic loss, propellant budget, wheel saturation.

The applicant's physical hypothesis is that the support force a body exerts on whatever holds it—its measured weight—is a constraint bill on the body's angular-momentum state rather than an immutable property, and that a body of ordered medium whose internal angular-momentum population is aligned and coherently driven at its precession resonance may exhibit a commandable change in that support force. Whether any cell exhibits such a response, with what sign and magnitude, is decided by measurement in a factory qualification stage; the systems of this application are built from cells that have passed that stage and operate from the resulting stored maps. Because the hypothesized lever acts on the constraint force itself rather than on momentum exchange with a surrounding medium, a responsive cell class would be matter- and atmosphere-independent: the same platform class would operate over ground, water, air, or in vacuum wherever a support or constraint force exists, and the customary boundaries between vehicle categories—drawn by the media they push against—would blur into a single multimodal class. This application claims the systems built from such cells and the methods of operating them, conditional on qualified cells operating within their maps.

No claim requires momentum creation ex nihilo. Every claimed system trims the support force on driven media it carries and pays the energy bill for any mechanical work through the cells' oscillating drive; the controller logs that bill. Within the hypothesis the cell alters the exchangeable orientation and precession state of the medium's internal population, not the constitutive angular momentum the medium carries; the medium's inertial mass is therefore taken as unchanged, and only the support force is trimmed. That assumption is stated, not proven, and is one of the quantities the qualification stage records.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background teaching only; no material essential to enablement, written description, or the claims is incorporated from them.

SUMMARY

The disclosure is organized in three tiers (FIG. 11). The products and their deployed operating methods lead; the qualification stage establishes the maps the products run from; the validation, recalibration, and fail-safe provisions govern what a deployed system may command.

Tier 1 — the products and their deployed operating methods (leading). Every product is built from **weight-trim cells** (400; FIG. 1), each comprising a body of ordered medium on a force interface through which its support force passes to the structure the cell serves; an alignment-field source; a coupling structure and locked oscillating drive at the medium's located precession resonance; a reaction frame carrying the field source and coupling structure on a separate load path with a reaction-force witness; runtime witness sensors; a calibration memory holding the cell's **empirical response map** with at least one independently validated nonzero operating point, uncertainty, provenance, validity envelope, hold-time characteristic, runtime limits, and safe-off parameters; and a cell controller that commands a support-force setpoint only from that map. The products are: (A) **lift platforms** of N cells on a rigid frame with a system controller, external-reference force account, and measured array allocation map, and **transport vehicles** built on them (FIGS. 2–4); (B) **hoisting and elevator systems** in which cells act as commanded variable-weight ballast or load relief with brake interlocks (FIG. 5); (C) **launch-assist trim rings and station-keeping vehicles** (FIG. 6); (D) **aerospace momentum-management systems** producing sign-paired differential torque and phased collective coupling trim (FIG. 7); (E) **marine surface and subsurface trim systems** commanding draft, list, trim, and depth without ballast transfer (FIG. 10); and (F) **descent and soft-landing systems** (FIG. 8). A common control topology (FIG. 3), failure-mode architecture (FIG. 9), and lift budget (FIG. 12) apply across families.

Tier 2 — qualification. Each cell's map is established once, per unit or production class, by a null-valid qualification protocol on an independent force reference with randomized drive and control states, reaction-force accounting, magnetic-gradient mapping, absorbed-energy-matched controls, preregistered gates, and independent replication; an assembled plurality is qualified as an array to produce a **measured array allocation map**. A cell whose map contains only null points is a null reference and cannot be operated in setpoint mode. A **qualified cell class** is a production class whose class map contains validated nonzero points and whose members pass an acceptance test against a separate force reference.

Tier 3 — validation, recalibration, and fail-safe. Validity states; an **external-reference validation layer** that compares the sum of cell-reported offsets with an external force reference (landing-structure load cells, hook load cell, draft or depth sensor, inertial vertical acceleration, orbit determination) and de-rates or safes the system on mismatch; built-in self-tests; and two loss-of-drive postures: **passive-safe static support** for ground-, hook-, brake-, buoyancy-, and orbital-velocity-supported embodiments, and **managed loss of drive** (redundant cells, reserve energy, controlled drive-on descent, co-integrated decelerator, minimum-safe-altitude interlock) for embodiments that are unsupported at zero drive.

Two operating regimes (FIG. 12). With ΔW_i the commanded change in support force of cell i and $\phi_i = \Delta W_i / [(m_c + m_s) g]$ its offset ratio referred to the cell's own weight (medium plus mount), a system's net support-force margin is $\sum_i \Delta W_i - N (m_c + m_s) g$ relative to the weight of everything

else it carries. In the **self-null regime** ($|\phi| \leq 1$) a cell can at most cancel its own weight; benefit arises where the cell's mass *replaces* mass the system would carry anyway (ballast, counterweight, adapter structure, shielding) or where the load-bearing structure itself is the driven medium (**integrated-medium configuration**), and where both signs are available the cell supplies $2(m_c + m_s)$ g of authority per cell. In the **super-unity regime** ($\phi > 1$) the cell's support force is driven negative beyond its own weight; payload lift-off, propellant-free ascent, altitude station-keeping, and full-propulsion replacement require it and are disclosed as prophetic, conditional on a qualified cell class whose map covers such points. Neither regime is presumed reachable.

Layered scope. Every quantity is disclosed at three layers: (L0) the raw synchronous observable—the change in a force reading synchronized with a commanded drive state; (L1) the validated response—the map-corrected support-force change with uncertainty inside the validity envelope; and (L2) controlled operation—setpoint control of a system's net support force, attitude, rate, torque, draft, or depth from the maps. No layer presumes sign, magnitude, additivity, retained state, or scaling.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Weight-trim cell (400): ordered medium in holder on a force interface to the served structure; alignment-field source; coupling structure and locked pulsed drive chain on a reaction frame with reaction-force witness; optional vertical drive-transparent guide with distributed coupling and position sensing; runtime witness sensors; calibration memory holding the empirical response map; cell controller with safe-off; cell interface.
- FIG. 2 — Lift platform (490): N cells on a rigid frame; common alignment-field bus; power plant; drive-power distribution; thermal management; landing structure with external-reference load cells; system controller with array allocation memory.
- FIG. 3 — Control topology: per-cell setpoint controllers; attitude loop (IMU, differential allocation) and collective loop (altimeter or external force reference); array allocation map; external-reference validation comparator; validity-state gate feeding safe-off or managed loss of drive.
- FIG. 4 — Transport unit: level platform with payload area, guidance, power plant, landing gear or skirt with load cells; translation by ground-contact drive wheels at reduced normal load, auxiliary electric thrusters, or differential support; conditional body-axis species noted.
- FIG. 5 — Hoisting and elevator systems: crane with integrated-medium hook block and hook load cell; elevator car with variable-weight ballast cells, drive machine, brake, counterweight (reduced, retained, or omitted by regime); brake interlock on loss of lock or validity.
- FIG. 6 — Launch and station-keeping: (a) assist configuration, payload-adapter trim ring on a launch stack with staging controller and boost-segment timeline; (b) full configuration (prophetic), vehicle whose cells offset its entire weight, with power plant sized to $M g v/\eta$, reserve energy, and managed loss of drive.

- FIG. 7 — Aerospace momentum management: sign-paired modules at $\pm r$ producing pure torque about axes perpendicular to the local vertical; third-axis actuator; reaction wheels desaturated; collective coupling trim phased over the orbit (timeline) for orbit-energy pumping and drag makeup.
- FIG. 8 — Descent and soft-landing: descending body with cells, descent controller, co-integrated decelerator, touchdown force sensor; drive-power-vs-altitude profile; regime note (atmosphere: terminal-velocity reduction; vacuum: super-unity required).
- FIG. 9 — Failure-mode ladder: cell dropout, lock loss, power loss, sign fault, and external-reference mismatch, each mapped to a mitigation; terminal posture branches into passive-safe static support (supported embodiments) and managed loss of drive (unsupported embodiments).
- FIG. 10 — Marine embodiments: surface vessel with draft sensors, inclinometer, cells replacing ballast, collective and differential loops; subsurface vehicle with depth sensor, sign-selected depth command, hull-compression compensation; passive-safe to the certified light-ship condition.
- FIG. 11 — Three-tier architecture and layered scope: factory qualification producing the cell map and array allocation map $\rightarrow$ qualified cell class or null reference $\rightarrow$ deployed operation with runtime validation $\rightarrow$ recalibration loop; layers L0, L1, L2 with the gates between them.
- FIG. 12 — Lift budget and regimes: net support force at the landing structure versus offset ratio ϕ for a representative platform; self-null and super-unity regimes; the integrated-medium configuration; lift-off threshold $\phi_{\text{lift}} = 1 + W_{\text{other}} / [N (m_c + m_s) g]$.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names without committing the specification or claims to any entity ontology associated with them. Every measured claim quantity is fixed by a recited instrument reading.

1. **Ordered medium; internal angular-momentum population** — a body whose constituent angular momenta (extant label: "electron spin," measured in aggregate as magnetization, A/m) can be aligned by a static field. Reference species: single-crystal yttrium iron garnet (YIG); alternatives are qualified on their own measurements.
2. **Alignment field; alignment sense** — the static bias field (oersted or tesla) applied to establish a measured alignment state of the population; the sense is its polarity. The support-force sign associated with either sense is taken from the map, never from the polarity.

3. **Precession resonance; locked drive; drive lock** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance), located operationally as the absorption extremum; a drive is *locked* when the cell controller's servo holds the drive frequency at that extremum within a stored tolerance; *loss of lock* is departure beyond the tolerance.
4. **Oscillating drive; resonant illumination** — a time-periodic drive applied to the medium through a coupling structure at the located precession resonance, specified by frequency, delivered power, pulse timing, and duty at the stated port; a medium is *resonantly illuminated* while the coupling structure delivers locked drive to it at or above the map's minimum absorbed power.
5. **Drive state** — the recorded operating variables for one observation: field magnitude and sense, drive frequency and phase, incident and absorbed power, pulse width, repetition rate, duty, resonance mode, alignment state, temperature, orientation, mounting class, and time.
6. **Weight-trim cell (cell; module)** — the apparatus of FIG. 1: an ordered medium and holder on a force interface; alignment-field source; coupling structure and locked drive chain on a reaction frame with a separate load path; runtime witness sensors; a calibration memory holding an empirical response map; and a cell controller executing the deployed operating method. "Module" is used interchangeably, especially for sealed cells.
7. **Trimmed quantity; support force of the driven medium** — the force transmitted through the cell's force interface by the driven medium and its mount to the structure the cell serves (newtons, positive downward along the local vertical when supported). Where the structure is supported (ground, hook, brake, buoyancy), it is the medium's share of the structure's measured weight. Where the structure is in free fall (orbit, ballistic descent), no support force exists, and the trimmed quantity is the **gravitational coupling** of the medium to the primary body—the term that, within the hypothesis, fixes the medium's natural-orbit radius—expressed as the equivalent support force the medium would exert if held at rest at that radius. The two definitions coincide wherever a support exists. The trimmed quantity is never "the scale reading" of the whole system; it is the medium's own contribution, and a system-level effect follows by ordinary statics only because the medium is rigidly borne.
8. **Commanded offset ΔW_i ; offset ratio ϕ_i** — ΔW_i is the commanded change in the trimmed quantity of cell i relative to its drive-off baseline (newtons; positive = lighter). The offset ratio is $\phi_i = \Delta W_i / [(m_c + m_s) g]$, with m_c the medium mass, m_s the mass of the cell's mount, housing, and any structure share assigned to the cell, and g the local gravitational acceleration (in the rotating frame of a rotating primary, the constraint acceleration, i.e., gravitational minus centrifugal).
9. **Self-null regime; super-unity regime** — $|\phi| \leq 1$ and $\phi > 1$ respectively (Definition 8). A negative ϕ (heavier) is available only where the map records it.
10. **Integrated-medium configuration** — a configuration in which a structural, ballast, counterweight, adapter, shielding, or hook-block mass that the system carries in any case is itself the ordered medium of one or more cells, so that its weight is commandable without adding mass.

11. **Net support-force margin** — for a system carrying N cells and other mass of weight W_{other} , the quantity $\sum_i \Delta W_i - N (m_c + m_s) g - W_{\text{other}}$; the system's measured weight on its support is $-(\text{margin})$ while that value is positive, and the system leaves its support when the margin exceeds zero. **Lift-off threshold:** $\phi_{\text{lift}} = 1 + W_{\text{other}} / [N (m_c + m_s) g]$ for uniform cells.
12. **Empirical response map (stored)** — the data structure held in a cell's calibration memory: the change in the trimmed quantity relative to the drive-off baseline over a set of drive states, each with uncertainty, hysteresis and time constants, thermal, orientation, and mounting-class coefficients, the **hold-time characteristic** (Definition 13), the identity of every operating point independently validated under the qualification gates, the provenance record, validity envelope and permissible interpolation region, runtime limits, and safe-off parameters. A map may contain only null points. It presumes no sign, magnitude, additivity, retained state, or scaling beyond its validated points.
13. **Hold-time characteristic** — the measured decay of the trimmed quantity after drive termination (time constant τ_h) and the measured drive duty D_h required to sustain a given offset within a stated ripple. Where $\tau_h \gg$ the pulse period the offset persists between pulses and the structure sees a sustained force; where it does not, the structure sees a pulsed force of duty D and the map so records. No value of τ_h is presumed.
14. **Provenance record** — unit or class identity; qualification apparatus, force references, reference bodies used; date; operator or process; raw-data identifier; fit method; replication record; gates passed for each validated point.
15. **Validity envelope; permissible interpolation region; runtime limits** — the region of drive-state and environmental variables (temperature, orientation, mounting class, force-axis field gradient, vibration and acceleration, absorbed power and duty, bandwidth and slew, drift rate, zero-check interval) over which the map was established and within which interpolation between validated points is permitted. Extrapolation is not permitted.
16. **Safe-off** — the stored state the cell controller enters on a validity failure, fault, or command: oscillating drive off; alignment field to a stored safe magnitude and sense (or unchanged in a fixed-bias cell); last valid record preserved; recalibration flag set. At safe-off the trimmed quantity returns to its drive-off baseline within τ_h .
17. **Qualified cell; qualified cell class; null reference** — a cell (or a production class represented by qualified units) whose map contains at least one independently validated nonzero operating point with uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters is a *qualified cell (class)*; its setpoint mode is enabled. A cell whose map contains only null points is a *null reference*: it accepts test-state commands and reports L0/L1 quantities but has no setpoint mode. A member of a qualified class is accepted under the class map only after an acceptance test at one or more validated points against a separate force reference agrees within a stored tolerance.
18. **Array allocation map** — for a plurality of cells on a common structure, the measured relation between sets of per-cell drive states and the change in the structure's system-level support force (or

torque, for differential sets), established under sequential, in-phase, phase-scrambled, and inactive-cell qualification states, with uncertainty and cross-talk terms, together with the hold power and, where climb is qualified, the offset-to-work efficiency η measured at array qualification. Additivity is not presumed; differential setpoints for attitude or torque are commanded only from differential responses recorded in it.

19. **External force reference; external-reference validation** — a force or motion reading on a load path outside every cell (landing-structure or landing-gear load cells; hook or car load cell; draft, displacement, or depth sensor; inertial vertical acceleration; orbit determination) against which the system controller compares the allocation-map prediction of $\Sigma \Delta W_i$; mismatch beyond a stored tolerance transitions the system to degraded, de-rate, or safe-off.
20. **Validity states** — *valid* (setpoint mode permitted), *degraded* (offsets held or reduced; update required), *out-of-envelope* (a witness beyond a runtime limit; setpoint inhibited), *invalid* (map check, gate, or external-reference failure; recalibration flagged), and *safe-off*.
21. **Passive-safe (static support)** — the property of a system that, at zero drive, remains statically supported by landing structure, hook and brake, buoyancy, or the orbital velocity it already carries. It is asserted only of embodiments having such a support.
22. **Managed loss of drive** — for embodiments unsupported at zero drive (payload lift-off, ascent, altitude station-keeping, orbit command by collective trim, vacuum descent deceleration), the subsystem and behavior of FIG. 9 branch (407): redundant cells with re-allocation, reserve energy sufficient for a controlled drive-on descent from the current state to touchdown or to a supported state, a co-integrated decelerator, and a minimum-safe-altitude/energy interlock that inhibits commands the reserve cannot cover.
23. **Body-axis response (conditional species)** — a hypothesized change in the force on the medium along a cell-fixed axis not aligned with the local vertical. It is not the trimmed quantity of Definition 7; it is available to a system only if a cell class's map records it as a separately validated response with its own gates. No embodiment of this application requires it.
24. **Natural-orbit radius; gravitational coupling (hypothesis terms)** — within the hypothesis, a free body's force-free path is set by its angular-momentum state and its coupling to the primary; changing the medium's driven state is hypothesized to change that coupling and thereby the radius at which the body is force-free. These terms label the hypothesis; every claim quantity is a force, torque, rate, draft, depth, or orbital element read by a recited instrument.

THE WEIGHT-TRIM CELL (400) — REPRESENTATIVE SPECIES (FIG. 1)

The cell is described here to the extent the systems require; it is the actuator every family is built from, and a system controller treats it as a calibrated force-offset source with a stored map.

Ordered medium (410) and holder (436). The reference medium is single-crystal YIG: a polished sphere of 2–15 mm diameter (reference 10 mm; volume 0.5236 cm³; mass $\approx$ 2.71 g at 5.17 g/cm³; drive-off support force $\approx$ 26.6 mN at standard gravity) for actuator-class cells, or a tiled or mosaic assembly of

such elements, boules, or shaped elements sharing one alignment field and one or more coupling structures for kilogram-class and larger cells. A tiled assembly is a single cell only if it is qualified as one; otherwise each element is a cell with its own map. Substituted garnets and engineered media are alternatives, each qualified on its own measurements. The holder is low-loss and nonmagnetic (quartz, sapphire, PTFE, alumina) and forms the upper element of the force interface.

Force interface (426). The holder transmits the medium's support force to the served structure—frame, hook block, car, adapter, hull, or descending body—so that a change in the medium's support force appears as an equal change in the structure's support force by statics. Mounting class and orientation are drive-state variables; a map is established or confirmed in the mounting class in which the cell is deployed. Where the system requires it (lift-class cells in super-unity operation), the holder rides a **vertical drive-transparent guide (425)**—a low-friction, low-loss track (sapphire or PTFE-lined nonmagnetic rail or flexure) that constrains the medium laterally, permits vertical travel within the coupling structure, and is transparent to the drive—and **distributed coupling (430)** (segmented or traveling coupling elements whose overlap is mapped through the travel range) maintains resonant illumination through that travel; optical position sensing (460) reads the medium's position on the guide. The guide and containment double as drive shields (404).

Alignment-field source (440). A permanent-magnet yoke with trim winding (deployed reference; sense selection or fixed bias) or an electromagnet, producing 1500–2500 Oe at the reference medium, with shims for the gradient budget below; a common field bus (492) may supply the trim windings of all cells on a platform. Field magnitude and sense are recorded drive-state variables.

Coupling structure (420) and locked pulsed drive chain (450). A TE-mode rectangular cavity (reference TE₁₀₂ near 5.6 GHz for the reference medium), loop-gap, re-entrant, waveguide, or coil structure places the medium at a mapped transverse microwave magnetic-field antinode orthogonal to the alignment field. The chain comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), a pulse and burst modulator with a controller-enforced duty bound, a power amplifier sized to the map, a directional coupler, and incident, reflected, and transmitted power detectors. The controller servo-locks to the located resonance (Definition 3). Pulses are nonoverlapping; higher delivered energy is bursts with recovery intervals or thermally matched continuous-wave operation; $Q \times$ source power is not a power multiplier. Thermal design target class for a trim-class cell: pulsed operation near 1 % duty with watts-class average dissipation per cell; this is a design target, not a measured value, and duty and power scale to the thermal plant up to continuous operation where the map covers it.

Reaction frame (422) and reaction-force witness (424). The field source, coupling structure, feeds, thermal hardware, and shields are carried on a frame whose loads pass to the served structure through a load path separate from the force interface. A reaction-force sensor on that path is a runtime witness: a commanded change in the medium's support force that is accounted for within tolerance by an equal and opposite change on the reaction path is an internal actuator artifact (magnetic force between source and medium, radiation pressure, thermal or acoustic force, guide stiction) and the cell reports *invalid*. Medium displacement on the guide is a position reading, not by itself evidence of a support-force change.

Runtime witness sensors (428). Medium and mount temperature; orientation; a force-axis field-gradient witness; acceleration and vibration; incident, reflected, and absorbed drive power and duty; lock state; reaction-frame force; and, where present, a force sensor in the force interface. Gradient budget at runtime: for a saturated reference sphere (moment $\approx 0.074 \text{ A} \cdot \text{m}^2$) a force-axis gradient of $\approx 1.32 \text{ } \mu\text{T/m}$ imitates 98 nN; the runtime limit keeps the residual gradient uncertainty below a preregistered fraction (preferably one third) of the gradient that would imitate the smallest reportable offset.

Calibration memory (432). Holds the empirical response map (Definition 12), provenance, validity envelope, runtime limits, safe-off parameters, unit or class identifier, validity-state history, zero-check history, and recalibration schedule. The controller refuses a map whose provenance does not match the identifier.

Cell controller (470) and safe-off (434). Receives a support-force setpoint; verifies it lies within the permissible interpolation region and inhibits it otherwise; inverts or interpolates the map to a drive state; commands the field source and drive chain within the stored slew and bandwidth limits; regulates by feedforward from the map and, where a force sensor is present in the interface, by feedback; compares witnesses with runtime limits on every control cycle; reports achieved offset with propagated uncertainty and validity state; and executes safe-off (Definition 16). No sign, magnitude, or additivity is computed from a theoretical coefficient.

Cell interface (480). Mechanical: the force interface and reaction path. Electrical: drive and field power. Data: support-force setpoint or test-state command in; achieved offset, uncertainty, position, lock state, validity state, temperature, hold-time status, and the cell's permissible region and runtime limits out.

Deployed operating method of a cell. (a) Load map; verify provenance; drive-off zero check (and sense-reversed check where permitted); enter *valid* only on pass. (b) Receive setpoint from the system controller. (c) Check region and runtime limits. (d) Determine mapped drive state. (e) Apply within slew limits. (f) Regulate. (g) Monitor witnesses and the reaction path. (h) Report. (i) Interleave zero checks at the stored interval. (j) Safe-off on exceedance, lock loss, or fault. (k) Recalibrate via a reduced re-qualification or return to full qualification. The operator of a deployed system does not perform qualification.

QUALIFICATION, ARRAY QUALIFICATION, AND THE QUALIFIED CELL CLASS (FIG. 11, Tier 2)

Cell qualification. Once per unit or production class, the medium in its deployment mounting class is placed on an independent sample force sensor with the field source, coupling structure, feeds, and shields on a separately supported frame carrying a reaction-force sensor; the controller executes a randomized, optionally blinded sequence of field-off, field-on/drive-off, resonant-drive, absorbed-energy-matched detuned or phase-scrambled drive, reference-body, position-swap, sense-reversal, and positive-control injection states; synchronized logs include support force, reaction force, drive power, temperature, pressure, vibration, acceleration, strain, field, and gradient. Acceptance gates, all preregistered: **Q1** repeatability across blocks and days; **Q2** resonance contrast against the absorbed-energy-matched control;

Q3 material contrast against matched reference bodies and swaps; **Q4** magnetic-force exclusion by the mapped gradient acting on the measured moment; **Q5** thermal, radiometric, buoyancy, and convection exclusion (pressure sweep); **Q6** mechanical, electrical, and reaction-path exclusion; **Q7** statistical validity after search and multiple-comparison correction; **Q8** independent replication on a second apparatus, unit, or laboratory. Only a Q1–Q8 point is a validated operating point. The hold-time characteristic is measured by drive-termination transients on the force channel within its bandwidth, separated from cavity ring-down and thermal and mechanical response. Output: the map, and the status *qualified cell* or *null reference*.

Production class. A class map is established from a plurality of qualified units of common design; an additional unit is accepted under it only after an acceptance test at one or more validated points against a separate force reference agrees within a stored tolerance.

Array qualification. The assembled plurality on its frame, hook block, car, adapter, or hull section is driven from a common phase reference through per-cell trim under sequential, in-phase, phase-scrambled, and inactive-cell states with per-cell and system-level force and reaction channels; additivity, cancellation, coupling-mode splitting, cross-talk, and differential (torque) responses are measured and written to the array allocation map (Definition 18). In the super-unity branch the array is qualified in containment (guide, position sensing, tether or captive rig) with incremental extension of drive authority inside the validated envelope; unsupported lift is tested only after the measured net margin approaches zero without a failed gate.

What qualification does not presume. Sign, magnitude, ϕ reachable, linearity with population or power, additivity, retained state, τ_h , lift-off, or any power-to-offset law; each is an output.

LIFT BUDGET AND REGIMES (FIG. 12)

For a structure of other weight $W_{\text{other}} = W_{\text{frame}} + W_{\text{payload}}$ (everything that is not a cell) carrying N cells each of own weight $(m_c + m_s) g$ and commanded offset ΔW_i :

measured weight at the support $W_{\text{sys}} = W_{\text{other}} + \sum_i [(m_c + m_s) g - \Delta W_i] = W_{\text{other}} + N (m_c + m_s) g (1 - \phi)$ (uniform ϕ)

lift-off condition $\sum_i \Delta W_i - N (m_c + m_s) g \geq W_{\text{other}} \Leftrightarrow \phi \geq \phi_{\text{lift}} = 1 + W_{\text{other}} / [N (m_c + m_s) g]$.

At $\phi = 1$ the cells are weight-neutral and the structure still weighs W_{other} ; it is not buoyant. The add-on benefit of a cell of mass m is $(\phi - 1) m g$: negative at zero drive (dead mass), zero at $\phi = 1$, positive only in the super-unity regime. In the integrated-medium configuration the medium is mass the system carries anyway, so W_{other} shrinks by the medium's weight and the self-null regime yields up to 100 % of that mass's weight; with both signs available a cell of mass m supplies $2 m g$ of vertical authority (from $+m g$ heavier to $-m g$ lighter), which is the basis of the ballast and dynamic-balancing embodiments. Inertial mass is taken as unchanged (Background), so for a system leaving its support the net upward acceleration is $a = [\sum \Delta W_i - N (m_c + m_s) g - W_{\text{other}}] / M_{\text{total}}$.

Any mechanical work raised through the cells is drawn from the drive: raising M_{total} at climb rate v requires drive throughput $P \geq M_{\text{total}} g v / \eta$, η the offset-to-work efficiency measured at array qualification and stored in the array allocation map (an output, not presumed), distinct from hold dissipation. The hold power to sustain a given ϕ is likewise an allocation-map output.

DETAILED DESCRIPTION OF THE SYSTEMS (Tier 1)

B1 — Lift platform (490; FIG. 2). A rigid frame carrying N cells ($N \geq 3$ for attitude authority; reference $N = 4, 6, 8$) with force interfaces to the frame and cell axes substantially parallel to the local vertical; a common alignment-field bus (492); power plant (494: battery, generator, or tether) sized to the hold power and, where climb is intended, to $M_{\text{total}} g v / \eta$; drive-power distribution (493); thermal management (495); landing structure (468) with **external-reference load cells (462)**; a system calibration memory (463) holding the array allocation map; and a system controller (496). Every cell is a qualified cell in its deployment mounting class; the platform is operated only at allocation-map points. Regimes: (i) *supported trim* (self-null): the platform remains on its landing structure and its measured weight, attitude loading, and ground-contact normal force are commanded; (ii) *integrated-medium platform*: deck, frame members, or ballast are the medium; (iii) *super-unity lift* (prophetic): where the class map covers $\phi > \phi_{\text{lift}}$, the platform leaves its support and the collective loop commands vertical rate; managed loss of drive (B9) is mandatory in this regime.

B2 — Control topology and validation (FIG. 3). Three nested loops: per-cell setpoint controllers (497) executing the cell method; an attitude loop (498) from an inertial measurement unit (452) commanding differential allocation only from differential responses recorded in the allocation map (456); and a collective loop (499) commanding net support force from the external force reference (462) while supported, or altitude and climb rate from an altimeter (454) once unsupported, through the collective mapper (458). The **external-reference validation comparator (465)** compares the allocation-map prediction of $\Sigma \Delta W_i$ against the load-cell sum (or, unsupported, against inertial vertical acceleration and altimeter rate): agreement within tolerance keeps *valid*; mismatch transitions to *degraded* and de-rate, then *safe-off* or managed loss of drive. **Built-in self-test:** sign-reversal test (commanded $+\Delta W$ then $-\Delta W$ must produce equal and opposite external-reference change where both signs are mapped); a **reference cell** (identical cell with a non-alignable medium) whose reported offset must read null; drive-on/drive-off differencing at the stored interval; the reaction-path test of every cell. Air-mass, ion-wind, and convective artifacts are absent in vacuum and in water and are bounded in air by the pressure-sweep record of qualification; a platform in air runs the self-test in a sealed skirt state where the design permits.

B3 — Transport unit (FIG. 4). The platform of B1 with a payload area (469), guidance system (464), power plant, and landing gear or skirt (468) with load cells. The platform is held level by the attitude loop. **Translation** is by ground-contact drive wheels or tracks (466) operating at a commanded reduced normal load (rolling resistance and grade force scale with the trimmed normal load while traction is preserved by allocation), by auxiliary electric thrusters (467), or by differential support on a skirt; none of these relies on tilting the platform. A change in the trimmed quantity is along the local vertical (Definition

7) and has no horizontal component in any attitude; tilt-vectoring is therefore not an embodiment of this application unless a cell class's map records a body-axis response (Definition 23), in which case, and only then, the controller may project that separately validated response by the recorded geometry. Design target class: low-clearance translation with wheels unloaded to a commanded fraction, not open-sky flight; free translation without ground contact is the super-unity regime.

B4 — Hoisting, crane, and elevator systems (FIG. 5). *Crane/hoist:* a load-bearing member (cable, hook block 474, spreader; crane structure 476; load 472) carries one or more cells whose force interface lies in the load path; the **hook block or spreader is itself the medium** (integrated-medium configuration) so that the self-null regime relieves up to the block's own weight, and add-on relief of load weight beyond that is the super-unity regime. A hook load cell (485) is the external reference; the controller cross-checks it against the achieved offset every cycle. Interlocks: trim ramp-rate limit; brake (482) engagement on loss of lock, invalid state, or cross-check mismatch; slew limited so the brake can arrest the unrelieved load. *Elevator:* cells (474) on the car frame (478) act as **commanded variable-weight ballast**: with both signs mapped, cells of total mass m_b provide $\pm m_b g$ about their own weight, so the controller balances car + payload against the counterweight (484) exactly at any payload, and the drive machine (483) supplies only acceleration and friction—a real machine-rating reduction in the self-null regime (P2). Counterweight reduction or omission beyond that balance is the super-unity regime. Regenerative descent is conventional; net grid draw equals drive energy plus losses. Brake on loss of lock or validity; the brake is rated for the undriven imbalance.

B5 — Launch and station-keeping (FIG. 6). *(a) Assist configuration.* A payload-adaptor trim ring (486) or strap-on set on a launch stack (488), with a staging controller (442) and power source, commands an offset during a selected boost segment and powers down or jettisons after it. The benefit is $\Delta v = \phi \mu g t$ for vertical flight (an upper bound during a gravity turn, $g \sin \gamma$), where $\mu = m_{\text{trim}} / m_{\text{stack}}$ is the trimmed mass fraction and the add-on ring's own weight is charged: net benefit $(\phi - 1) (m_{\text{ring}} / m_{\text{stack}}) g t$ for an add-on ring, or $\phi (m_{\text{adapter}} / m_{\text{stack}}) g t$ where the adapter structure is the medium. Inertial (thrust) loads are unchanged; only the gravitational share of adapter load and gravity loss is reduced. *(b) Full configuration (prophetic).* A vehicle (487) whose cells' net margin exceeds its other weight ascends at a commanded rate with no rocket propulsion, powered by an onboard plant (489) or tether sized to $M_{\text{total}} g v / \eta$ plus hold power, and holds station at altitude by continued commanded offset. In the rotating frame of the primary the required offset at station is the constraint acceleration (gravitational minus centrifugal; 0.41 % less than gravitational at 400 km over the equator). Within the hypothesis a body whose coupling is so trimmed is force-free where it stands; this is not represented as achieved. Managed loss of drive (B9) is mandatory; the plant, reserve energy, and decelerator are part of M_{total} and are themselves trimmed.

B6 — Aerospace momentum and attitude management (FIG. 7). Body-mounted or integrated cells (446) on a vehicle (444) in the natural-orbit field of a primary body—near-surface through orbital class. The trimmed quantity here is the gravitational coupling of each medium (Definition 7), commandable in sign where mapped. *Differential torque:* cells at $\pm r$ from the center of mass commanded to **sign-paired offsets** ($+\Delta W$ at $+r$, $-\Delta W$ at $-r$) produce a pure torque $2 \Delta W r$ about an axis perpendicular to both r and

the local vertical with no net force; single-sign differential sets produce torque plus a net force and are used only where that force is wanted. Because every offset is along the local vertical (the radial constraint gradient, 449), torque is available about the two axes perpendicular to it (the gravity-gradient geometry); the **third axis (about the local vertical) is served by a magnetorquer or small wheel (447)**, or by a body-axis response if a class map records one. Assist configurations desaturate reaction wheels (448); full replacement of wheels and control-moment gyroscopes is two-axis by cells plus third-axis actuator. *Collective coupling trim*: a collective offset ϕ at fixed velocity places the vehicle at periapsis of an orbit of eccentricity $e \approx \phi$; **phasing** the offset over the orbit—lighter (reduced coupling) on the outbound half where the radial velocity is positive, restored on the inbound half—adds orbital energy at the drive's expense, $\Delta E \approx 2 a e \phi m g$ per orbit, $\Delta a \approx 4 a e \phi$ per orbit for near-circular orbits (P5), providing drag makeup and orbit raising; the reverse phasing lowers. Orbit determination is the external reference against which the controller validates the achieved coupling trim. At zero drive the vehicle retains the orbit its velocity supports (passive-safe only in that sense); orbit-command maneuvers are managed-loss-of-drive operations: the controller commands no phasing whose interruption would leave a perigee below a stored minimum.

B7 — Marine surface and subsurface vehicles (FIG. 10). Net vertical force = buoyancy – measured weight; both signs of offset are useful and the self-null regime suffices. *Surface vessel (510)* at waterline (512): cells (400) sized and placed to replace variable ballast—cells of total mass m_b spanning $-m_b g$ to $+m_b g$ replace $2 m_b$ of ballast capacity with no water moved, no pumps, no ballast-water treatment; the controller commands draft (514/516) from draft sensors (517) through the collective loop and list and trim (515) from an inclinometer (518) through the differential loop; cargo-exchange compensation is a setpoint change. *Subsurface vehicle (520)*: depth command from a depth sensor (519): heavier to descend (522), lighter to ascend (524), null to hold (526), with hull-compression buoyancy loss compensated within the map; emergency ascent by full trim toward weight null; hull-load relief at depth by fractional offset. Marine controller (528). Passive-safe: total drive loss restores the vessel to the **certified light-ship-plus-fixed-cell-mass condition** for which stability and propeller immersion are certified; the cells' own zero-drive weight is part of the certified condition. Sealed marine housings; water excludes air-mass artifacts.

B8 — Descent control and soft landing (FIG. 8). Cells on a descending body (412), a descent controller (414) profiling drive power against altitude and descent rate, a touchdown force sensor (418), and a **co-integrated decelerator (416)** (parachute, airbag, or crushable structure). In atmosphere, trimming the body's weight to $(1 - \phi_{\text{eff}})$ of baseline reduces terminal velocity by $\sqrt{1 - \phi_{\text{eff}}}$ and touchdown load accordingly; this works in the self-null regime for the medium's share and requires super-unity for the remainder of the body. In vacuum a body already falling is decelerated only by a negative net weight (super-unity). Loss-of-drive posture: descent at zero drive is no worse than the baseline undriven descent with the decelerator; the controller deploys the decelerator on validity loss above a stored altitude.

B9 — Failure modes and safety (FIG. 9). Cell dropout → re-allocation among remaining cells from the allocation map (401); lock loss → re-acquisition by the cell servo within a stored window, else safe-off of that cell (402); power loss or brownout → graceful collective de-rate (403); drive containment (guides

and housings double as shields) (404); field-sense interlock preventing uncommanded sign reversal (405); external-reference mismatch → de-rate then safe-off (465). Terminal posture: **passive-safe static support (406)** for platforms on landing structure, hoists and elevators (brake), marine vessels (buoyancy), and orbital vehicles carrying orbital velocity; **managed loss of drive (407)** for unsupported embodiments—redundant cells sized so the loss of any one leaves the allocation map able to cover the current setpoint, reserve energy (408) sufficient for a controlled drive-on descent from the current altitude and rate to touchdown or to a supported state, co-integrated decelerator (416), and a minimum-safe-altitude/energy interlock (409) that inhibits any ascent, station, or orbit-phasing command the reserve cannot cover. No embodiment asserts passive safety it does not have.

PROPHETIC EXAMPLES (prospective; no result achieved; every number follows from the stated masses, $g = 9.80665 \text{ m/s}^2$, and the stated assumptions; response maps are assumed for the paper example only)

P1 — platform sizing (both regimes). Platform of $N = 8$ uniform cells, each $m_c = 60 \text{ kg}$ medium (tiled assembly) + $m_s = 40 \text{ kg}$ mount, so $(m_c + m_s) g = 980.7 \text{ N}$ per cell; frame 2,000 kg; payload 1,000 kg; $W_{\text{other}} = 29,420 \text{ N}$; $M_{\text{total}} = 3,800 \text{ kg}$. Hold power $P_c(\phi)$ and efficiency η are map outputs and are not predicted.

ϕ	$\Sigma \Delta W_i \text{ (N)}$	$W_{\text{sys at landing structure (N)}}$	state
0	0	37,265	undriven; $W_{\text{sys}} = W_{\text{other}} + 8 \times 980.7$
0.001	7.8	37,257	actuator-class trim
0.01	78	37,187	trim-class
0.1	785	36,481	2.1 % of platform weight relieved
1.0	7,845	29,420	cells weight-neutral; platform weighs W_{other}
2.0	15,691	21,575	super-unity; still supported
4.75	37,265	0	$\phi_{\text{lift}} = 1 + 29,420/7,845$
6.0	47,072	−9,807 (net upward)	$a = 9,807/3,800 = 2.58 \text{ m/s}^2$

Integrated-medium variant: if the 2,000 kg frame is itself the medium of the eight cells ($m_c = 250 \text{ kg}$, $m_s = 0$), $W_{\text{other}} = 9,807 \text{ N}$ (payload only), $N(m_c) g = 19,613 \text{ N}$, and $\phi_{\text{lift}} = 1.5$; at $\phi = 1$ the platform weighs only its payload. Climb at $v = 1 \text{ m/s}$ of the 3,800 kg platform requires drive throughput $P \geq 37.3 \text{ kW}/\eta$.

P2 — elevator dynamic balancing (self-null regime). Car 500 kg; rated payload 400 kg; cells of total mass $m_b = 200 \text{ kg}$ on the car with both signs mapped (authority $\pm 1,961 \text{ N}$); counterweight set to 900 kg = car + cells + half payload. Conventional practice (counterweight = car + half payload) leaves a static imbalance of up to $\pm 200 \text{ kg} \times g = \pm 1,961 \text{ N}$. With cells the controller nulls that imbalance at any payload, so the machine supplies only acceleration and friction: at 1.0 m/s^2 on the 2,000 kg moving mass, peak machine force $\approx 2,000 \text{ N}$ versus $\approx 3,560 \text{ N}$ for the conventional car (1,961 N imbalance + 1.0 m/s^2 on

1,600 kg), a $\approx 44\%$ reduction in peak machine force in this example, at a 200 kg increase in car mass. Brake rated for the undriven imbalance ($\pm 200 \text{ kg} \times g$ here, the cells at their zero-drive weight). Counterweight omission is super-unity ($\phi > 1 + 900/200$ at full payload) and is not asserted.

P3 — launch and ascent ladder. Stack 10,000 kg; boost segment $t = 120 \text{ s}$; $v_e = 3.4 \text{ km/s}$; the effective stack fraction $\phi\mu$ governs: $\Delta v = \phi \mu g t$ (vertical), propellant relief $1 - e^{(-\Delta v/v_e)}$.

- $\phi\mu = 0.01$: $\Delta v = 11.8 \text{ m/s}$; relief 0.35 %. Realized by a ring of $m_{\text{ring}}/m_{\text{stack}} = 0.005$ at $\phi = 3$ (net $(\phi-1)\mu = 0.01$) or an integrated adapter of $\mu = 0.01$ at $\phi = 1$.
- Integrated adapter, $\mu = 0.02$, $\phi = 1$: $\Delta v = 23.5 \text{ m/s}$; 0.69 %.
- Add-on ring 500 kg ($\mu = 0.05$) at $\phi = 3$: net $\Delta v = 2 \times 0.05 \times g \times 120 = 117.7 \text{ m/s}$; 3.4 %.
- $\phi\mu = 0.3$: $\Delta v = 353 \text{ m/s}$; relief 9.9 % (exact), stage-resizing class; realized by $\phi = 0.3$ where the whole stack is medium ($\mu = 1$), by $\phi = 3$ on an integrated fraction $\mu = 0.1$, or by an add-on ring of $\mu = 0.1$ at $\phi = 4$ (net $(\phi-1)\mu = 0.3$).
- Full configuration, $\phi \geq \phi_{\text{lift}}$ for the whole vehicle: raising 10,000 kg through 400 km is $m g h = 3.92 \times 10^{10} \text{ J} = 10,900 \text{ kWh}$ at unit efficiency (true potential difference 10,270 kWh); the energy content of $\approx 0.91 \text{ t}$ of kerosene at 43 MJ/kg. Plant: a battery at 250 Wh/kg would mass $\approx 44 \text{ t}$ (exceeding the stack), so the full configuration presumes a generator or tether—at $\eta = 0.3$ overall, $\approx 3.0 \text{ t}$ of kerosene plus plant mass, all of it part of M_{total} and itself trimmed. Climb at 10 m/s needs throughput $\geq 0.98 \text{ MW}/\eta$; at 100 m/s, 9.8 MW/ η . Station at 400 km requires the constraint-acceleration offset (0.41 % below gravitational over the equator) held for the station duration at the map's hold power. Every rung uses the same cells and claims; the difference between rungs is ϕ and μ , i.e., the qualified class's map and how much of the vehicle is medium—materials and configuration variables, conditional on qualification. None is represented as reachable.

P4 — marine ballast replacement (self-null regime). Vessel of 10,000 t displacement, waterplane area 2,000 m², variable-ballast capacity 3,000 t. Cells of total mass 1,500 t with both signs mapped span -14.7 MN to $+14.7 \text{ MN}$ about their own weight, i.e., the same 3,000 t range of weight the ballast system provides, with no water moved; draft authority $3,000 / (1.025 \times 2,000) = 1.46 \text{ m}$. Zero drive leaves 1,500 t of fixed cell mass, which is the certified light-ship-plus-cells condition. List correction: differential allocation across port and starboard cells at the inclinometer's command.

P5 — orbital momentum management (self-null, small ϕ). Vehicle at 400 km ($a = 6,771 \text{ km}$, $g = 8.69 \text{ m/s}^2$) with two 100 kg cells at $r = \pm 5 \text{ m}$. Sign-paired $\phi = 10^{-4}$ ($\Delta W = 0.087 \text{ N}$ each) gives torque $2 \times 0.087 \times 5 = 0.87 \text{ N}\cdot\text{m}$ —above the 0.01–0.2 N·m class of small-satellite reaction wheels—with no net force. Collective phasing at $\phi = 10^{-3}$ on an orbit of $e = 10^{-3}$: $\Delta a \approx 4 a e \phi = 27 \text{ m}$ per orbit, $\approx 420 \text{ m/day}$ at 15.6 orbits/day, an order sufficient for drag makeup of a small vehicle at that altitude; the eccentricity is first induced by a collective step of the same ϕ . Orbit determination validates the achieved trim.

POAMS-NATIVE CELL SPECIES

The qualified-map cells are production species. A parallel prophetic species operates from the POAMS-native change in natural-orbit radius and constraint bill disclosed here and in the core family, with achieved authority measured during operation. Conventional translation or third-axis actuators remain optional fallbacks and do not exclude a separately qualified body-axis response.

CLAIMS

1. A lift platform system comprising: a rigid frame; a plurality of weight-trim cells fixed to the frame, each cell comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a holder on a force interface transmitting a support force of the body to the frame, an alignment-field source, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the aligned population, a reaction frame carrying the alignment-field source and the coupling structure and transmitting their loads to the frame through a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map relating a drive state of the cell to a change in the support force of the body, established by measurement of the cell or of a production class to which it belongs and containing at least one independently validated nonzero operating point, an uncertainty for each operating point, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller configured to command a drive state from the empirical response map for a support-force setpoint within the permissible interpolation region and to inhibit the setpoint otherwise; a power subsystem; a landing structure supporting the frame and carrying at least one external-reference force sensor reading a support force of the frame; a system calibration memory storing an array allocation map established by measurement of the assembled plurality of cells on the frame and relating sets of per-cell drive states to a change in the support force of the frame and to a differential response of the frame, with uncertainty; and a system controller configured to receive a net support-force setpoint and an attitude setpoint, to allocate per-cell support-force setpoints from the array allocation map within the permissible interpolation region of each cell, to compare a prediction of the array allocation map with the external-reference force sensor, to maintain a validity state, and to transition every cell to a safe-off state defined by its safe-off parameters when the comparison deviates beyond a stored tolerance or any cell reports an invalid state.
2. The system of claim 1, wherein each cell further comprises a vertical drive-transparent guide constraining the body laterally and permitting vertical travel within the coupling structure, a distributed coupling maintaining resonant illumination of the body through the travel, and a position sensor reading the body's position on the guide.
3. The system of claim 1, wherein the system controller closes an attitude loop by differential per-cell setpoints in response to an inertial measurement unit, the differential setpoints being allocated only from differential responses recorded in the array allocation map.

4. The system of claim 1, wherein a structural or ballast mass of the frame is itself the body of ordered medium of at least one of the cells.
5. The system of claim 1, wherein the system controller limits the sum of the commanded changes in support force to no more than the sum of the weights of the cells, whereby the frame remains supported by the landing structure at every commanded state and at zero drive.
6. The system of claim 1, wherein the array allocation map contains validated operating points at which the sum of the commanded changes in support force exceeds the sum of the weights of the cells plus the weight of the frame and a payload, the system controller is further configured to command a vertical rate of the frame by collective allocation from an altimeter and inertial vertical acceleration once the frame has left the landing structure, and the system further comprises a loss-of-drive management subsystem comprising redundant cells with re-allocation, reserve energy sufficient for a controlled drive-on descent to touchdown, a co-integrated decelerator, and a minimum-safe-altitude interlock inhibiting any command the reserve energy cannot cover.
7. The system of claim 1, wherein the empirical response map of each cell stores a hold-time characteristic comprising a decay time of the change in support force after drive termination and a drive duty required to sustain the change, and the cells are operated pulsed at a duty cycle of at most 5 % with a per-cell average dissipation of at most 10 W per kilogram of ordered medium in a hold regime in which no mechanical work is raised through the cells.
8. The system of claim 1, wherein the system controller re-allocates support-force setpoints among remaining cells from the array allocation map upon a cell reporting an invalid or safe-off state.
9. The system of claim 1, wherein the sum of the commanded changes in support force, less the sum of the weights of the cells, is staged at a fraction of the weight of the frame and a payload selected from: at least one percent; at least ten percent; at least fifty percent; and substantially the whole; each stage being commanded only where the array allocation map contains validated operating points covering it.
10. The system of claim 1, further comprising a built-in self-test comprising at least one of: a sign-reversal test in which equal and opposite support-force setpoints are commanded and the external-reference force sensor is required to register equal and opposite changes; a reference cell having a non-alignable body whose reported change is required to read null; and drive-on and drive-off differencing at a stored interval.
11. A method of operating a lift platform system having a frame, a plurality of weight-trim cells each holding an empirical response map established by a qualification procedure performed before the method and containing at least one independently validated nonzero operating point, a landing structure with an external-reference force sensor, and a system calibration memory holding an array allocation map established by measurement of the assembled cells, the method comprising: loading each cell's map and the array allocation map and verifying that each provenance record matches the cell or class and the assembled platform; executing a drive-off zero check against the

external-reference force sensor; receiving a net support-force setpoint and an attitude setpoint; allocating per-cell support-force setpoints from the array allocation map within each cell's permissible interpolation region and inhibiting any allocation outside it; applying, in each cell, the alignment field and the oscillating drive at the located precession resonance at the drive state its map assigns to its setpoint; comparing the array allocation map's predicted change in the frame's support force with the external-reference force sensor and, upon deviation beyond a stored tolerance, reducing the setpoints and, upon persistence, transitioning every cell to safe-off; commanding differential setpoints for attitude only from differential responses recorded in the array allocation map; and restoring the frame's support force by reducing the setpoints before the landing structure resumes full load; wherein the qualification procedure is not repeated during the method.

12. The method of claim 11, wherein the array allocation map covers a net support-force margin exceeding zero, the net support-force setpoint is commanded through zero so that the frame leaves the landing structure, a vertical rate of the frame is commanded by collective allocation from an altimeter and inertial vertical acceleration, and a loss-of-drive management subsystem is armed before the frame leaves the landing structure, the method inhibiting any ascent command whose controlled drive-on descent the subsystem's reserve energy cannot cover.
13. The method of claim 12, wherein all mechanical work performed in raising the frame is supplied through the cells' oscillating drive from a power subsystem whose throughput is at least the product of the total mass, the local gravitational acceleration, and the commanded vertical rate divided by an efficiency stored in the array allocation map, and drive termination restores the undriven support force within the stored hold time.
14. A transport vehicle comprising the system of claim 1, a payload area on the frame, a guidance system, and translation means selected from: ground-contact drive wheels or tracks operated at a normal load reduced by the commanded change in support force; auxiliary electric thrusters; and differential support on a skirt; wherein the system controller holds the frame level by differential allocation during translation and the change in support force is applied along the local vertical without tilting the frame.
15. The vehicle of claim 14, wherein at least one cell's empirical response map further contains a separately validated body-axis response along a cell-fixed axis not aligned with the local vertical, and the system controller projects that response by a geometry recorded in the map to supply a horizontal component of translation authority only where that response is so recorded.
16. A method of translating a payload comprising: supporting the payload on the vehicle of claim 14; commanding, from the array allocation map, a collective change in support force that reduces the normal load borne by the landing structure and the translation means to a selected fraction of the undriven normal load while the frame remains in ground contact; translating the vehicle by the translation means at the reduced normal load with the frame held level; and restoring the normal load by reducing the setpoints before stopping.

17. A hoisting system comprising: a load-bearing member; at least one weight-trim cell comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface transmitting a support force of the body into the load path of the load-bearing member, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame carrying the alignment-field source and the coupling structure on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and relating a drive state to a change in the support force of the body, the map containing at least one independently validated nonzero operating point, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; a load cell in the load path outside the cell reading the load; a brake acting on the load-bearing member; and a hoist controller configured to command a support-force setpoint of the cell within its permissible interpolation region, to cross-check the achieved change reported by the cell against the load cell on each control cycle, and to engage the brake upon loss of drive lock, an invalid state, or a cross-check mismatch beyond a stored tolerance, the brake being rated for the load at zero drive.
18. The hoisting system of claim 17, wherein a hook block or spreader of the load-bearing member is itself the body of ordered medium of the cell, whereby the commanded change in support force relieves up to the weight of the hook block or spreader without added mass.
19. An elevator comprising: a car; a drive machine with a brake; a counterweight; a plurality of weight-trim cells fixed to the car frame, each comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface transmitting a support force of the body into the car's load path, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and containing independently validated nonzero operating points of both signs with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; a car load cell; and an elevator controller configured to command the cells, from a measured allocation map of the cells on the car, to a net change in support force that balances the weight of the car, cells, and payload against the counterweight at any payload within the cells' authority, whereby the drive machine supplies only acceleration and friction, the controller engaging the brake upon loss of drive lock or an invalid state, the brake being rated for the undriven imbalance.
20. The elevator of claim 19, wherein the counterweight is reduced below the weight of the car and cells, or omitted, only where the measured allocation map contains validated operating points at which the net change in support force exceeds the weight of the cells by the resulting imbalance.

21. A method of operating a hoisting system or elevator having at least one weight-trim cell holding an empirical response map established before the method and containing at least one independently validated nonzero operating point, a load cell in the load path outside the cell, and a brake, the method comprising: reading the load on the load cell; commanding a support-force setpoint of the cell within its permissible interpolation region at a ramp rate not exceeding a stored limit; cross-checking the achieved change reported by the cell against the change read by the load cell on each control cycle; engaging the brake upon loss of drive lock, an invalid state, or a cross-check mismatch beyond a stored tolerance; and, where the system is an elevator having a car and a counterweight, adjusting the setpoint upon each change of payload so that the car and counterweight remain balanced within the cell's authority.
22. A launch-vehicle trim system comprising: a payload adapter carrying a plurality of weight-trim cells, each comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface transmitting a support force of the body into the adapter's load path, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and containing at least one independently validated nonzero operating point with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; a power source; and a staging controller configured to command the cells during a selected boost segment to a net change in support force within the cells' measured allocation map, to power the cells down or jettison them after the segment, and to record the commanded change together with the segment duration for accounting of the resulting velocity margin.
23. The trim system of claim 22, wherein the payload adapter structure is itself the body of ordered medium of the cells, whereby the commanded change relieves the gravitational share of the adapter's load and the gravity loss attributable to it without added mass.
24. A launch and station-keeping vehicle comprising the system of claim 1 wherein the array allocation map contains validated operating points at which the sum of the commanded changes in support force exceeds the total weight of the vehicle including its cells, power plant, and payload; a power plant whose throughput is at least the product of the vehicle's total mass, the local gravitational acceleration, and a commanded ascent rate divided by an efficiency stored in the array allocation map; and the loss-of-drive management subsystem of claim 6; the system controller being configured to command ascent at the commanded rate to a selected altitude and to hold station at that altitude by a continued commanded change equal to the vehicle's weight in the rotating frame of the primary body.

25. A method of assisting a launch comprising: carrying, on a payload adapter of a launch stack, weight-trim cells each holding an empirical response map established before the method and containing at least one independently validated nonzero operating point; during a selected boost segment, commanding the cells to a net change in support force within their measured allocation map; recording the commanded change and the segment duration; and powering the cells down or jettisoning them after the segment; wherein the velocity margin credited to the segment is the product of the commanded net change less the weight of any cells added to the stack, divided by the stack's weight, the local gravitational acceleration, and the segment duration.
26. A method of operating the vehicle of claim 24, comprising: arming the loss-of-drive management subsystem; commanding the net change in support force through the vehicle's total weight so that the vehicle leaves its support; commanding an ascent rate by collective allocation from an altimeter and inertial vertical acceleration while the power plant supplies the drive; on reaching the selected altitude, commanding a continued change equal to the vehicle's weight in the rotating frame of the primary body; and inhibiting any command whose controlled drive-on descent the reserve energy cannot cover.
27. An aerospace-vehicle momentum-management system comprising: a vehicle body; a plurality of weight-trim cells mounted to or integrated in the body at positions displaced from the body's center of mass along a body axis, each comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface rigidly joining the ordered medium to the vehicle body, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and recording changes, of both signs, in the gravitational coupling of the ordered medium to a primary body, with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; an attitude sensor; and a momentum controller configured to command sign-paired changes of equal magnitude and opposite sign to cells on opposite sides of the center of mass so as to produce a torque about an axis perpendicular to the body axis and to the local vertical with no net change in the vehicle's coupling, within each cell's permissible interpolation region.
28. The system of claim 27, further comprising reaction wheels, wherein the momentum controller commands the sign-paired changes to desaturate the reaction wheels.
29. The system of claim 27, further comprising a third-axis actuator selected from a magnetorquer and a momentum wheel providing torque about the local vertical, wherein attitude authority about the two axes perpendicular to the local vertical is provided entirely by the cells and the vehicle carries no reaction wheel or control-moment gyroscope for those two axes.

30. The system of claim 27, wherein the momentum controller is further configured to command a collective change in coupling to all cells phased with the vehicle's orbital position, reducing the coupling while the vehicle's radial velocity is positive and restoring it while the radial velocity is negative, so as to raise the orbit, or the reverse phasing so as to lower it, and to validate the achieved change against orbit determination, the controller inhibiting any phasing whose interruption would leave a perigee below a stored minimum.
31. A method of managing the momentum of an orbiting vehicle carrying weight-trim cells holding empirical response maps established before the method and recording changes in gravitational coupling of both signs, the method comprising: for attitude, commanding equal and opposite changes in coupling to cells on opposite sides of the vehicle's center of mass along a body axis so as to produce torque without net change in the vehicle's coupling; for orbit adjustment, commanding a collective change in coupling to all cells phased with the vehicle's radial velocity so as to add or remove orbital energy; comparing the achieved attitude and orbital elements with an attitude sensor and orbit determination and transitioning the cells to safe-off upon deviation beyond a stored tolerance; and inhibiting any phasing whose interruption would leave a perigee below a stored minimum.
32. A marine vessel comprising: a hull; a plurality of weight-trim cells fixed within the hull, each comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface transmitting a support force of the body into the hull's load path, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and containing independently validated nonzero operating points of both signs with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; at least one of a draft sensor and a depth sensor; an inclinometer; and a marine controller configured to command a collective change in support force from a measured allocation map of the cells in response to the draft or depth sensor to set draft or depth, and a differential change in response to the inclinometer to correct list and trim, without transfer of ballast water; the vessel at zero drive returning to a certified condition comprising the hull and the undriven weight of the cells.
33. The vessel of claim 32, wherein the cells are sized and placed so that the range of their commanded change in support force, from the heaviest to the lightest mapped state, equals the weight of ballast water the vessel would otherwise carry, the cells replacing the ballast system.
34. A method of controlling the depth of a subsurface vehicle carrying weight-trim cells holding empirical response maps established before the method and recording changes in support force of both signs, the method comprising: reading a depth sensor; commanding, from a measured allocation map, an increase in the cells' support force to descend, a decrease to ascend, and a

change that nulls the vehicle's net vertical force to hold depth, the change compensating the buoyancy loss from hull compression recorded in the map; reading an inclinometer and commanding a differential change for trim; and, upon loss of drive lock or an invalid state, transitioning the cells to safe-off so that the vehicle returns to its certified buoyancy condition; no ballast water being moved.

35. A method of controlling the draft and list of a surface vessel carrying weight-trim cells holding empirical response maps established before the method and recording changes in support force of both signs, the method comprising: reading a draft sensor and an inclinometer; commanding, from a measured allocation map, a collective change in the cells' support force to set a selected draft and a differential change to correct list and trim; and, upon a cargo exchange, commanding a change in support force equal to the change in cargo weight within the cells' authority; no ballast water being moved.
36. A descent-control system comprising: a descending body; a plurality of weight-trim cells fixed to the descending body, each comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, a force interface transmitting a support force of the ordered medium into the descending body's load path, an alignment-field source, a coupling structure and drive source applying an oscillating drive at a located precession resonance of the aligned population, a reaction frame on a load path separate from the force interface, runtime witness sensors, a calibration memory storing an empirical response map established by measurement and containing at least one independently validated nonzero operating point with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, and a cell controller commanding drive states only for setpoints within the permissible interpolation region; an altimeter and a descent-rate sensor; a co-integrated decelerator; a touchdown force sensor; and a descent controller configured to profile the cells' commanded change in support force against altitude and descent rate within a measured allocation map so as to reduce touchdown load, and to deploy the decelerator upon an invalid state or loss of drive above a stored altitude, whereby descent at zero drive is no worse than the undriven descent with the decelerator.
37. The system of claim 36, wherein the body descends in an atmosphere and the commanded change reduces the body's terminal velocity, or the body descends in vacuum and the measured allocation map contains validated operating points at which the sum of the commanded changes exceeds the total weight of the body.
38. A method of landing a descending body carrying weight-trim cells holding empirical response maps established before the method and containing at least one independently validated nonzero operating point, the method comprising: reading altitude and descent rate; commanding, from a measured allocation map, a change in the cells' support force profiled against altitude and descent

rate so as to reduce touchdown load; deploying a co-integrated decelerator upon an invalid state or loss of drive above a stored altitude; reading a touchdown force sensor; and reducing the setpoints to zero after touchdown.

39. The system, vehicle, elevator, or vessel of any one of claims 1, 14, 17, 19, 22, 24, 27, 32, and 36, wherein each cell further comprises a reaction-force sensor on its reaction load path and the cell controller reports the achieved change as invalid when it is accounted for within a stored tolerance by an equal and opposite change measured by the reaction-force sensor.
40. The system, vehicle, elevator, or vessel of any one of claims 1, 14, 17, 19, 22, 24, 27, 32, and 36, wherein the cells belong to a qualified production class whose class map contains the validated operating points and each cell was accepted under the class map by an acceptance test at one or more validated operating points against a separate force reference, and the measured allocation map was established by an array qualification operating the cells sequentially, simultaneously in phase, simultaneously with phase-scrambled references, and with selected cells inactive, without presuming additivity.
41. A system according to any one of claims 1, 14, 17, 19, 22, 24, 27, 32, and 36, wherein at least one weight-trim cell is a prophetic direct-operation cell configured to allocate driven angular momentum between intrinsic-turn and orbital-winding channels and to command the resulting natural-orbit-radius or support-force change without requiring a previously stored nonzero response map, the system controller measuring the achieved authority and enforcing the mass budget, axis limits, and loss-of-drive state otherwise recited.

ABSTRACT

Lift, transport, hoisting, launch, orbital momentum-management, marine trim, and soft-landing systems built from calibrated weight-trim cells, each cell comprising an ordered medium on a force interface, an alignment-field source, a locked resonant drive on a separate reaction path, a calibration memory holding an empirically established response map with at least one independently validated nonzero operating point, and a controller commanding support-force offsets only within that map. A system controller allocates per-cell offsets from a measured array allocation map, validates the sum against an external force reference (load cells, draft or depth sensor, inertial acceleration, or orbit determination), and enters safe-off on mismatch. In the self-null regime cells at most cancel their own weight and replace ballast, counterweight, or structure; payload lift-off, ascent, and station-keeping require the super-unity regime, disclosed as prophetic with managed loss of drive. Supported embodiments are passive-safe at zero drive. Methods of operating each system are claimed.

Weight-trim cell (400) — representative species

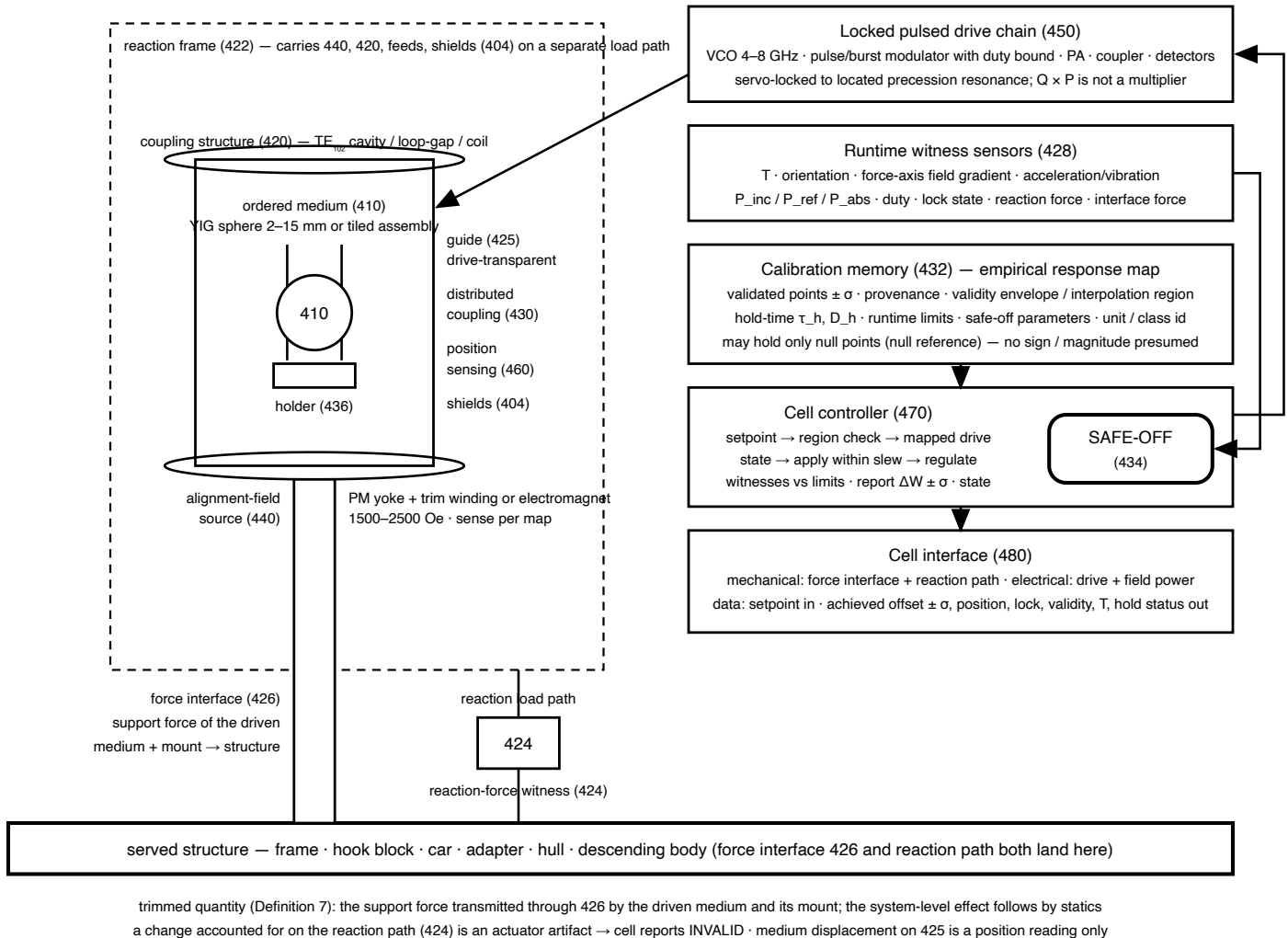


FIG. 1

Weight-trim cell (400) — ordered medium (410) in holder (436) on a force interface (426) to the served structure; alignment-field source (440); coupling structure (420) and locked pulsed drive chain (450) on a reaction frame (422) with reaction-force witness (424); optional vertical drive-transparent guide (425) with distributed coupling (430) and position sensing (460); runtime witness sensors (428); calibration memory (432) holding the empirical response map; cell controller (470) with safe-off (434); cell interface (480).

Lift platform — N weight-trim cells on a rigid frame (490)

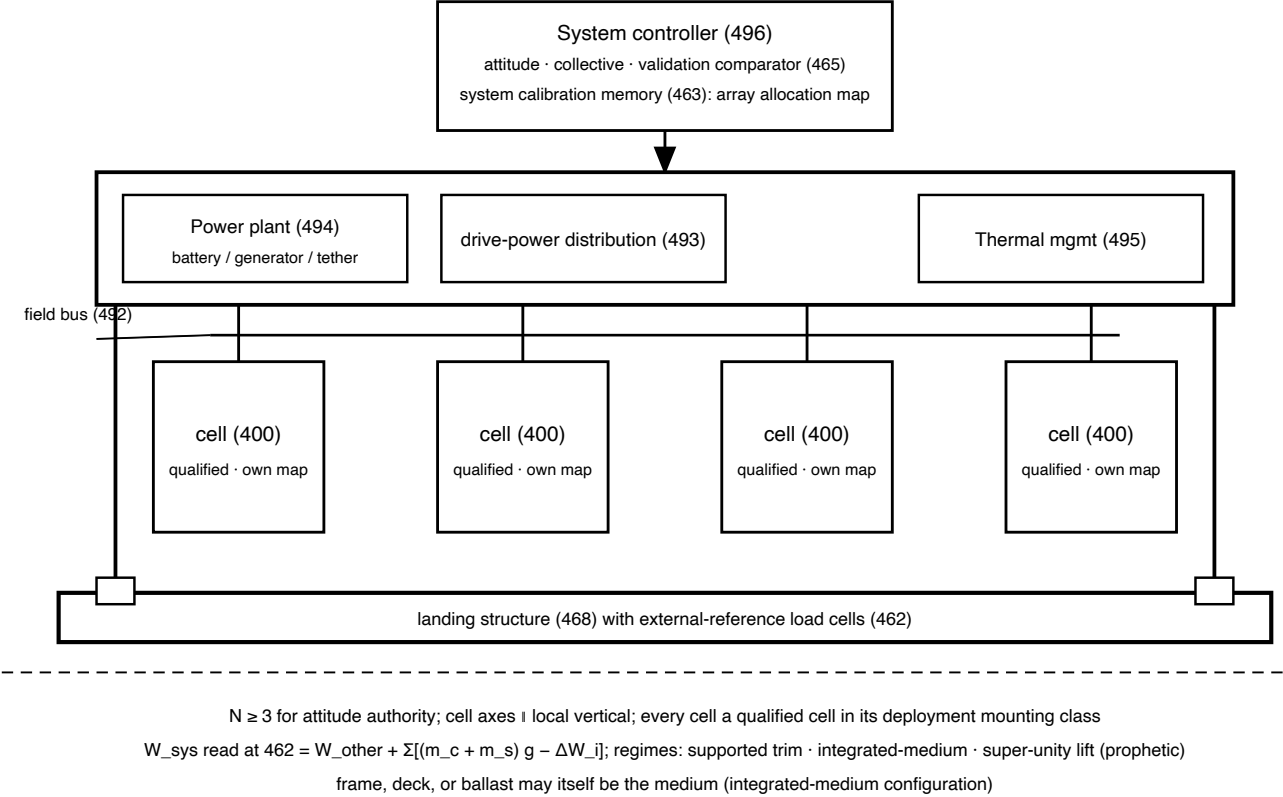
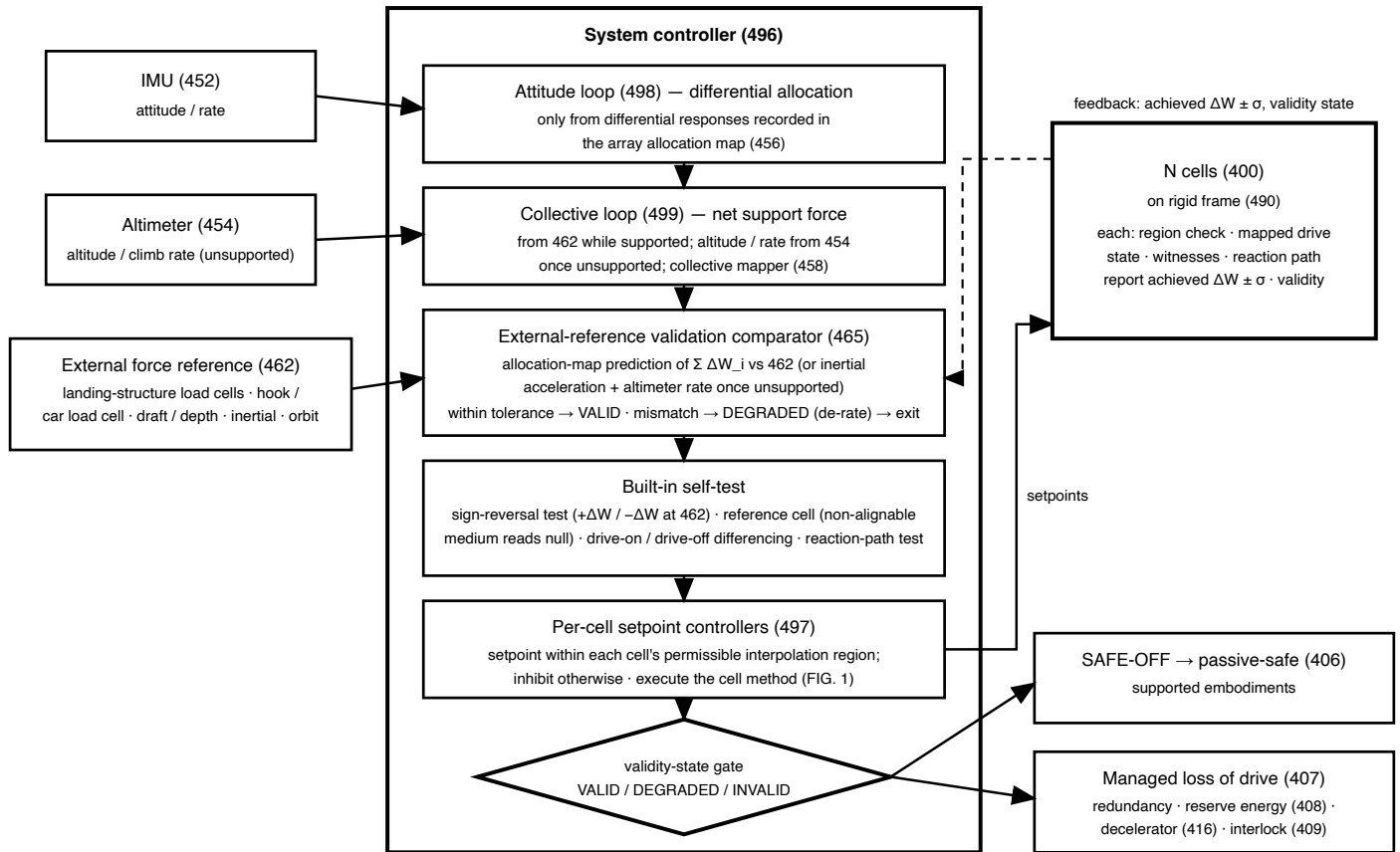


FIG. 2

Lift platform (490) — N cells on a rigid frame under a common alignment-field bus (492), power plant (494), drive-power distribution (493), thermal management (495); landing structure (468) with external-reference load cells (462); system controller (496) with system calibration memory (463) holding the array allocation map.

Control topology — per-cell setpoint controllers, attitude loop, collective loop, validation gate



no differential setpoint is commanded except from a recorded differential response; no setpoint outside a cell's permissible region;
mismatch between predicted $\Sigma \Delta W_i$ and the external reference de-rates, then safes the system — the posture depends on whether the embodiment is supported at zero drive

FIG. 3

Control topology — per-cell setpoint controllers (497); attitude loop (498) from the IMU (452) commanding differential allocation only from responses recorded in the array allocation map (456); collective loop (499) from the external force reference (462) or altimeter (454) through the collective mapper (458); external-reference validation comparator (465); built-in self-test; validity-state gate feeding safe-off / passive-safe static support (406) or managed loss of drive (407).

Transport unit — level platform, translation at reduced normal load

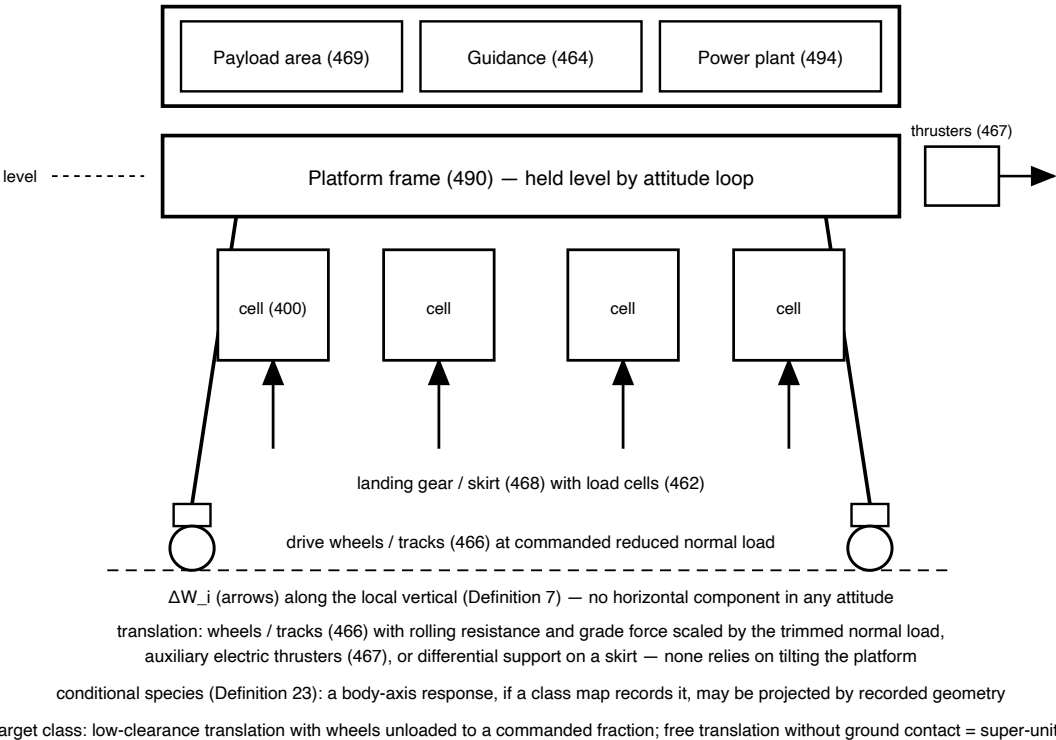


FIG. 4

Transport unit — level platform with payload area (469), guidance (464), power plant (494), landing gear or skirt (468) with load cells (462); translation by ground-contact drive wheels or tracks (466) at reduced normal load, auxiliary electric thrusters (467), or differential support; conditional body-axis species noted.

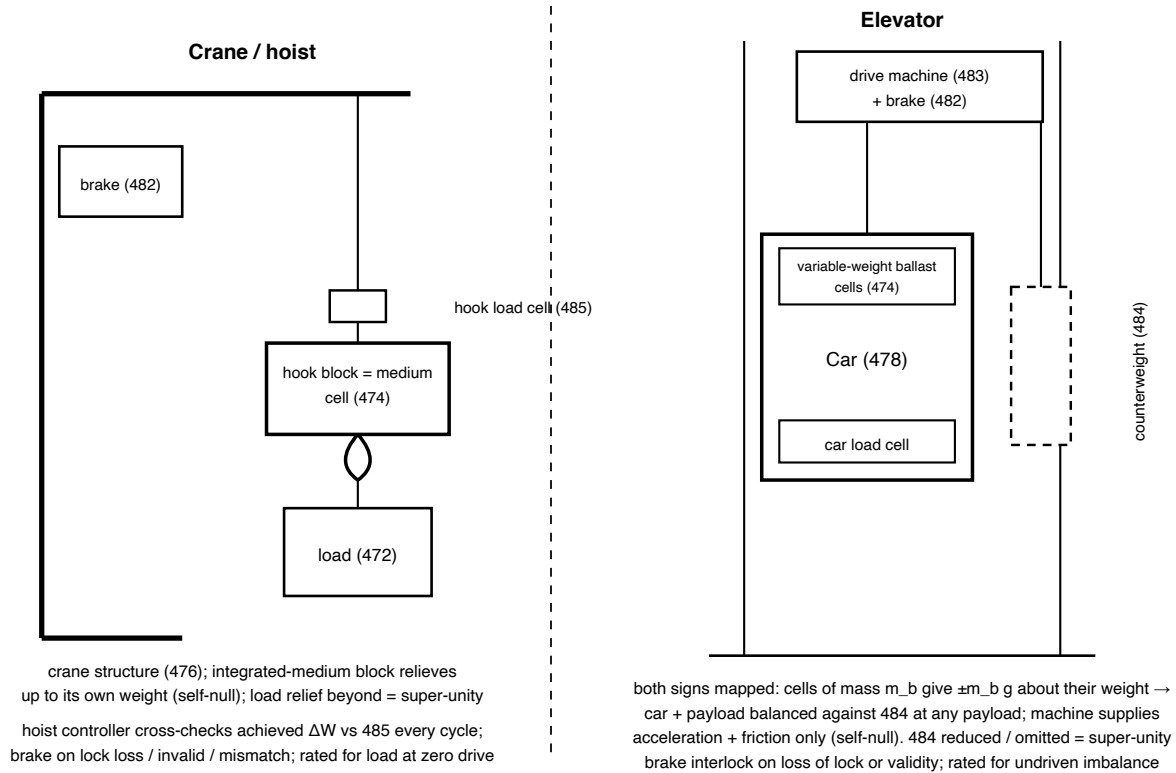


FIG. 5

Hoisting and elevator systems — crane with integrated-medium hook block (474), hook load cell (485), brake (482), crane structure (476), load (472); elevator car (478) with variable-weight ballast cells (474), drive machine (483), brake (482), counterweight (484) reduced, retained, or omitted by regime; brake interlock on loss of lock or validity.

Launch and station-keeping

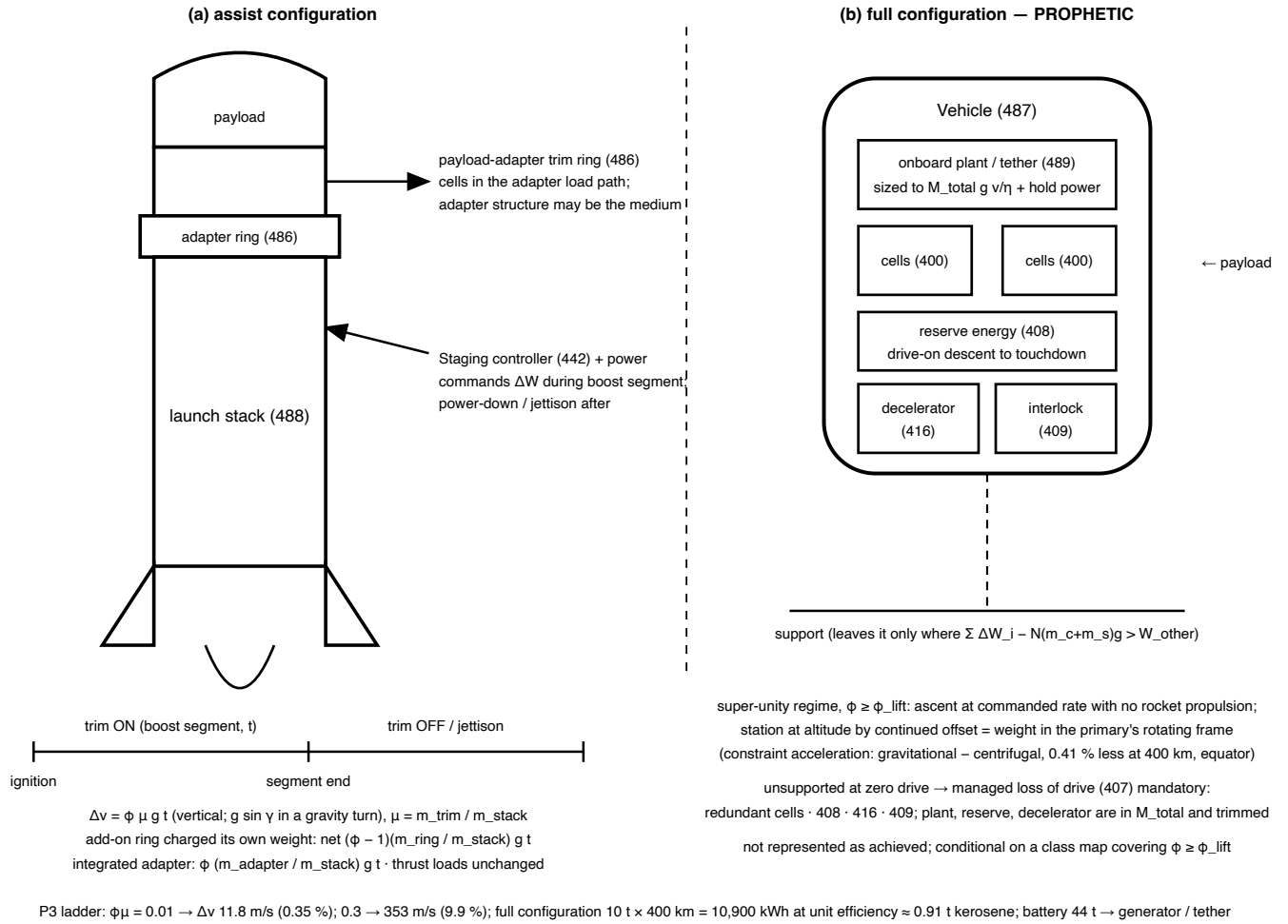


FIG. 6

Launch and station-keeping — (a) assist configuration: payload-adapter trim ring (486) on a launch stack (488) with staging controller (442) and boost-segment timeline; (b) full configuration (prophetic): vehicle (487) whose cells offset its entire weight, onboard plant (489) sized to $M g v / \eta$, reserve energy (408), decelerator (416), interlock (409), managed loss of drive.

Aerospace momentum management — sign-paired torque and phased collective coupling trim

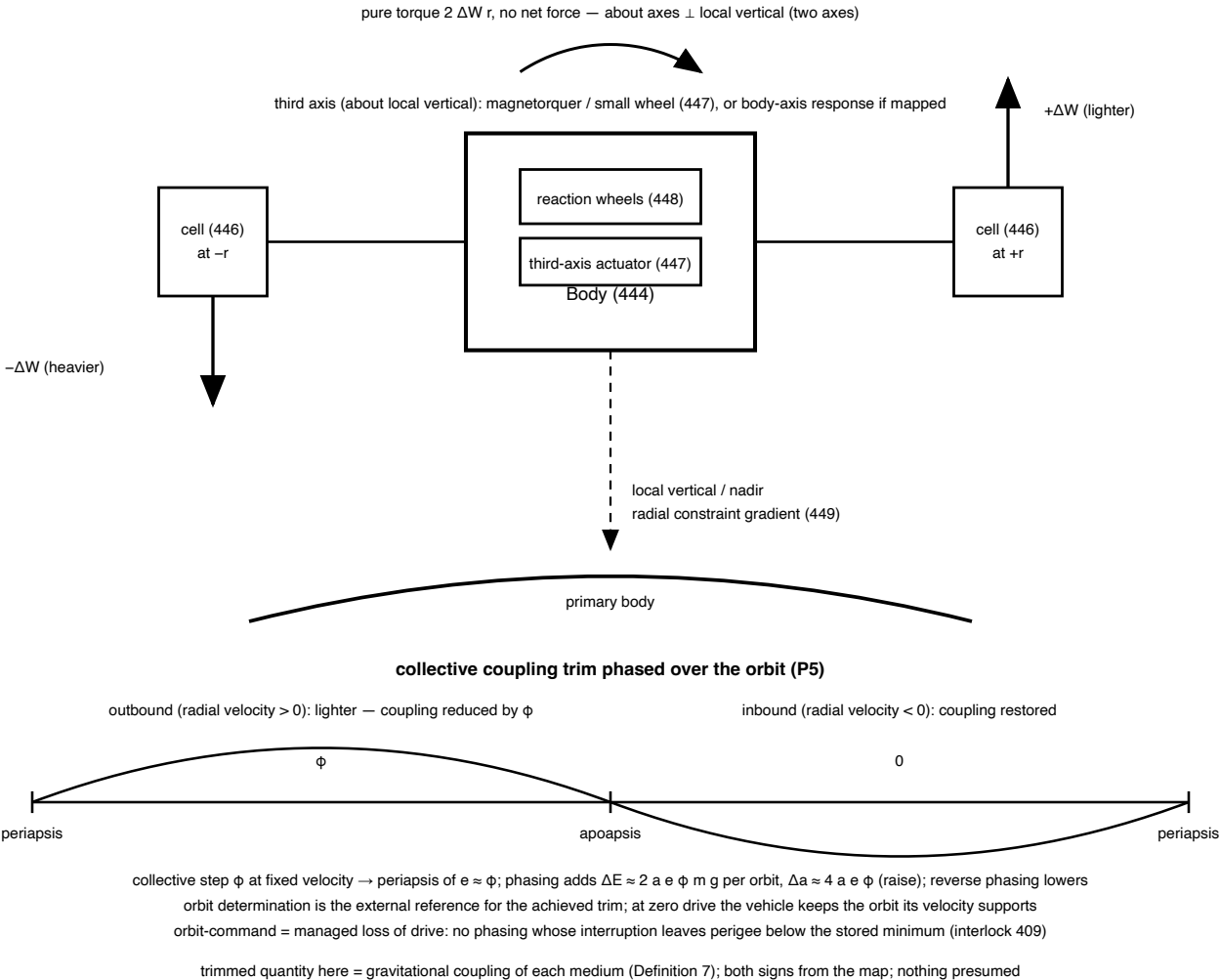
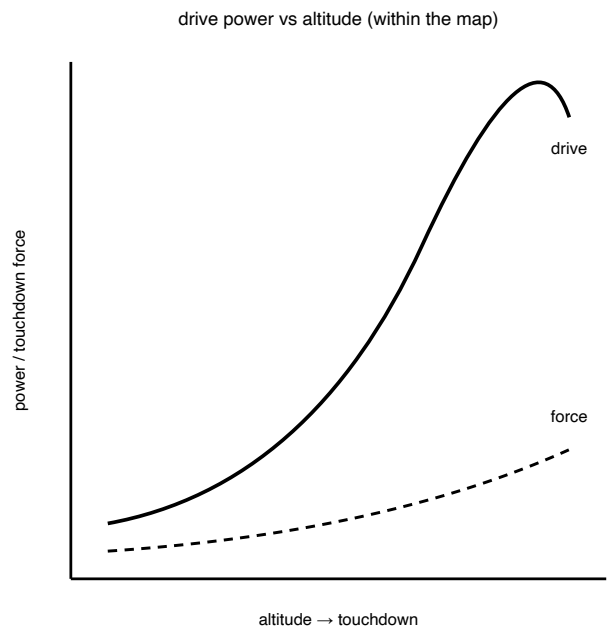
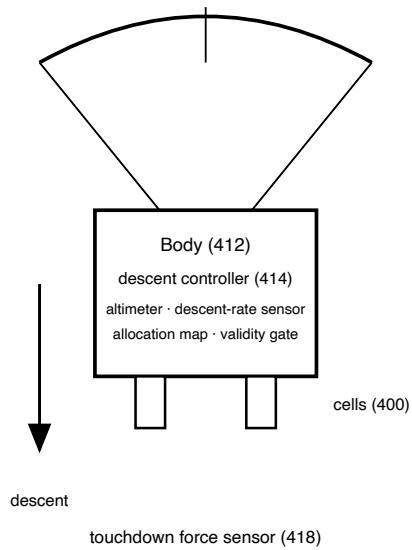


FIG. 7

Aerospace momentum management — body (444) with sign-paired cells (446) at $\pm r$ producing pure torque about axes perpendicular to the local vertical (radial constraint gradient 449); third-axis actuator (447); reaction wheels (448) desaturated; collective coupling trim phased over the orbit (timeline) for orbit-energy pumping and drag makeup.

Descent / soft landing

co-integrated decelerator (416): parachute / airbag / crushable



regime note — atmosphere: weight trimmed to $(1 - \phi_{\text{eff}})$ of baseline lowers terminal velocity by $\sqrt{1 - \phi_{\text{eff}}}$ and touchdown load; self-null covers the medium's share only, the remainder of the body needs super-unity. Vacuum: deceleration of a falling body requires super-unity.

loss-of-drive posture: descent at zero drive is no worse than the undriven descent with the decelerator (416); the controller deploys 416 on validity loss above a stored altitude; setpoints reduced to zero after touchdown (418)

no soft landing is represented as achieved; the profile is commanded only at allocation-map points

FIG. 8

Descent and soft landing — descending body (412) with cells (400), descent controller (414), co-integrated decelerator (416), touchdown force sensor (418); drive-power-vs-altitude profile; regime note (atmosphere: terminal-velocity reduction; vacuum: super-unity required); loss-of-drive posture no worse than the undriven descent.

Failure-mode ladder

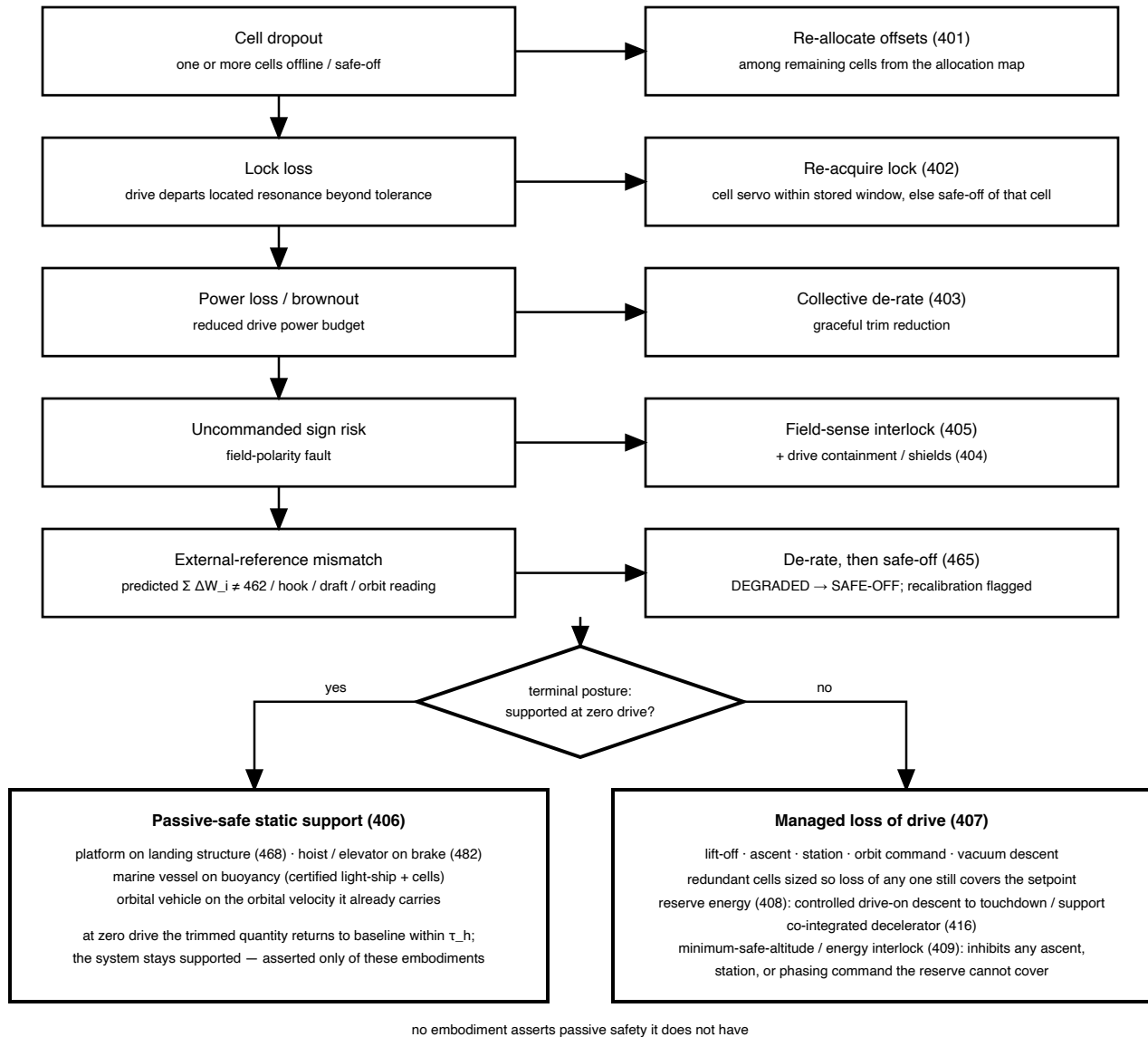


FIG. 9

Failure-mode ladder — cell dropout (401), lock loss (402), power loss (403), sign fault (404/405), and external-reference mismatch (465), each mapped to a mitigation; terminal posture branches into passive-safe static support (406) for supported embodiments and managed loss of drive (407) with reserve energy (408), decelerator (416), and interlock (409) for unsupported embodiments.

Marine embodiments — surface trim and subsurface depth command without ballast transfer

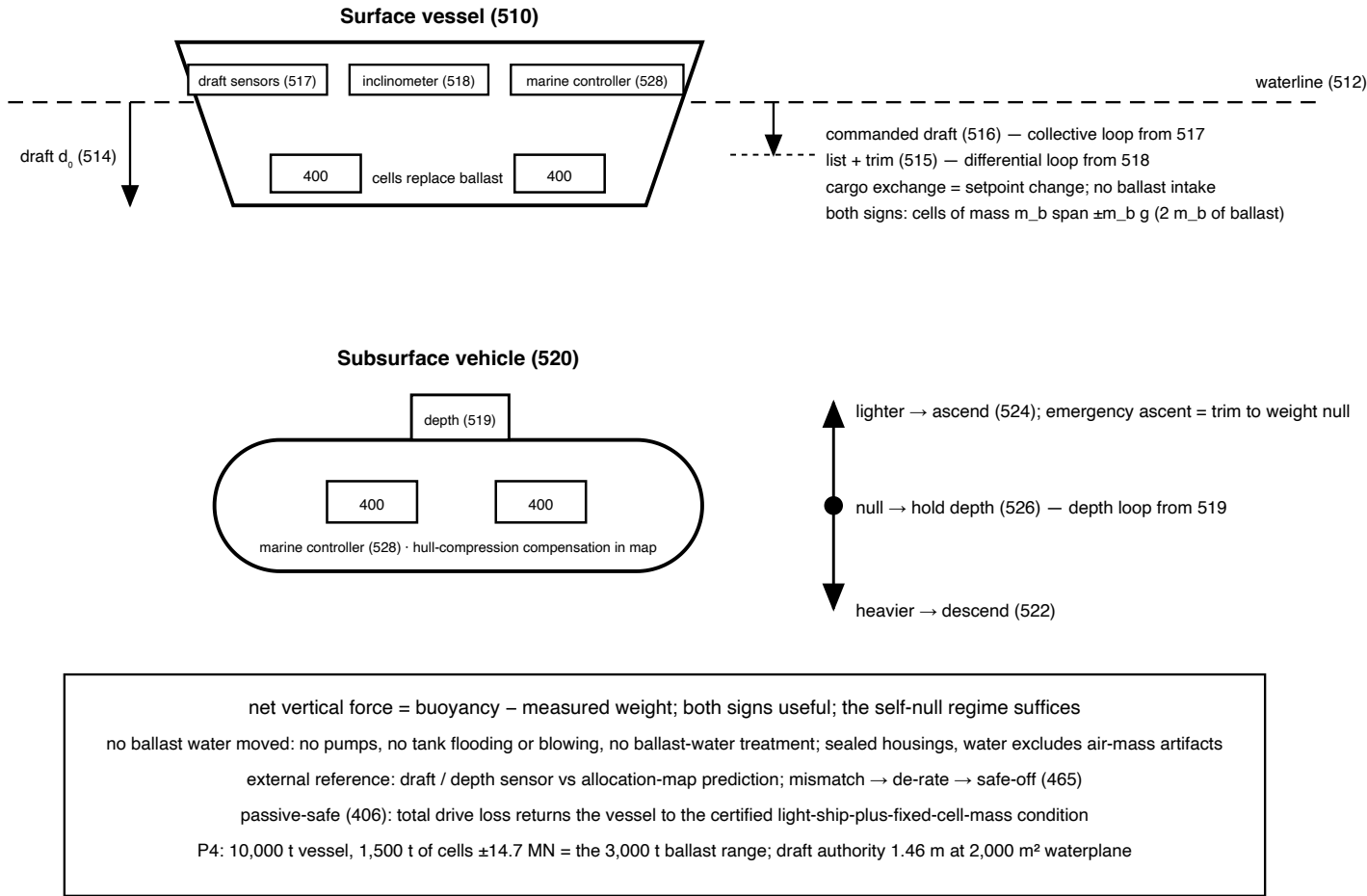
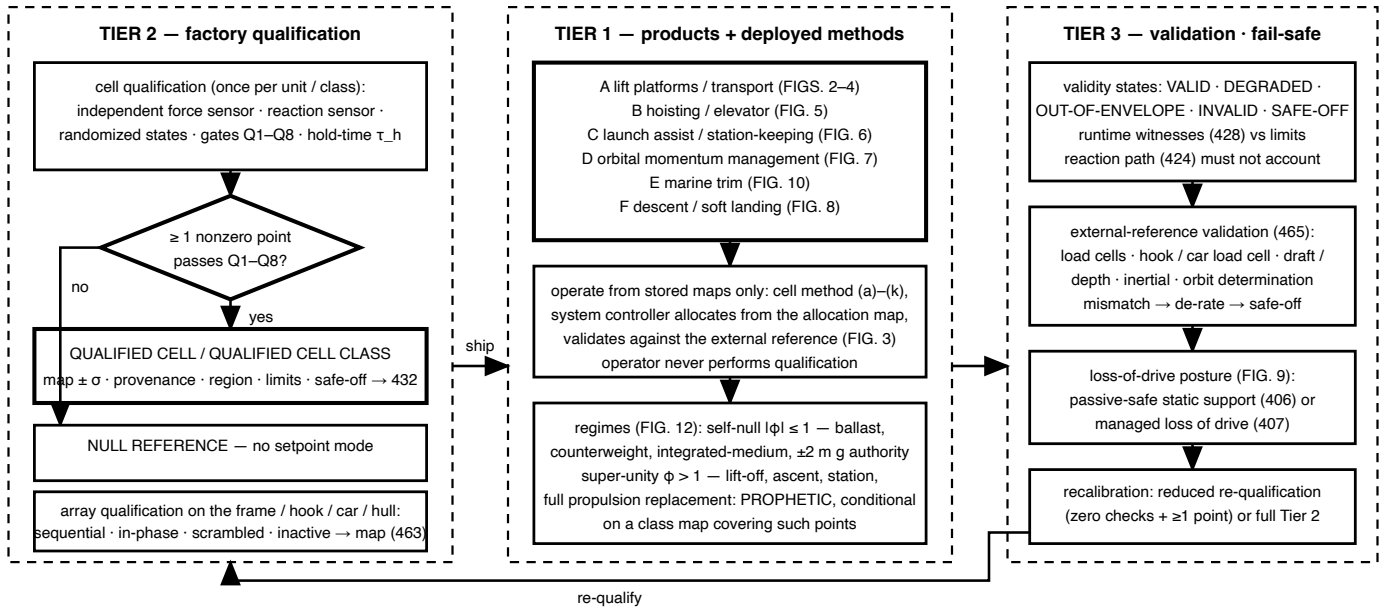


FIG. 10

Marine embodiments — surface vessel (510) with draft sensors (517), inclinometer (518), cells replacing ballast, collective (514/516) and differential (515) loops; subsurface vehicle (520) with depth sensor (519), sign-selected depth command (522/524/526), hull-compression compensation, marine controller (528); passive-safe to the certified light-ship condition.

Three-tier architecture and layered scope



Layered scope — every quantity disclosed at three layers; no layer presumes sign, magnitude, additivity, retained state, or scaling

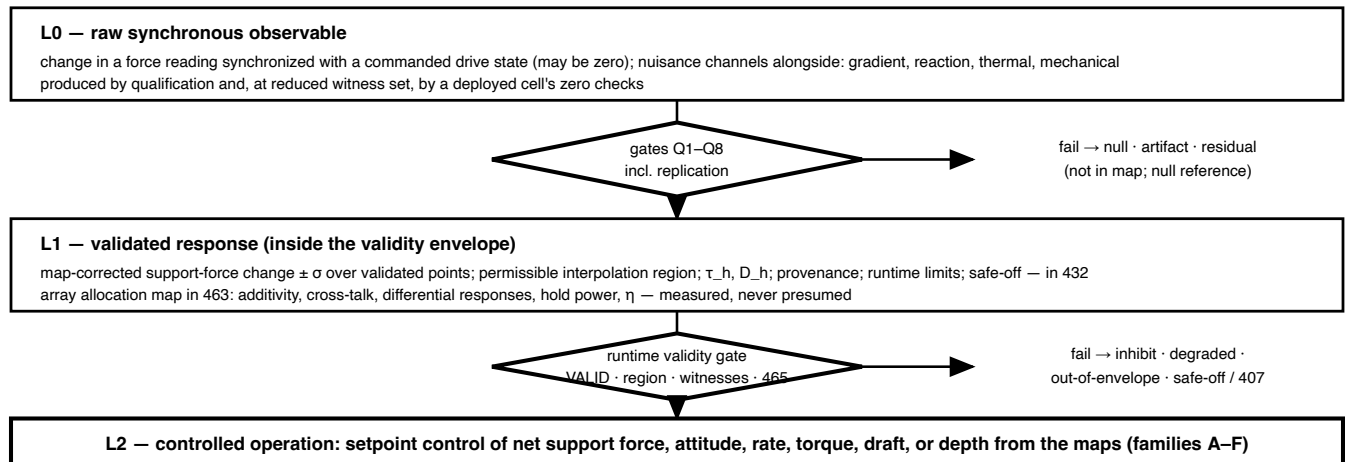
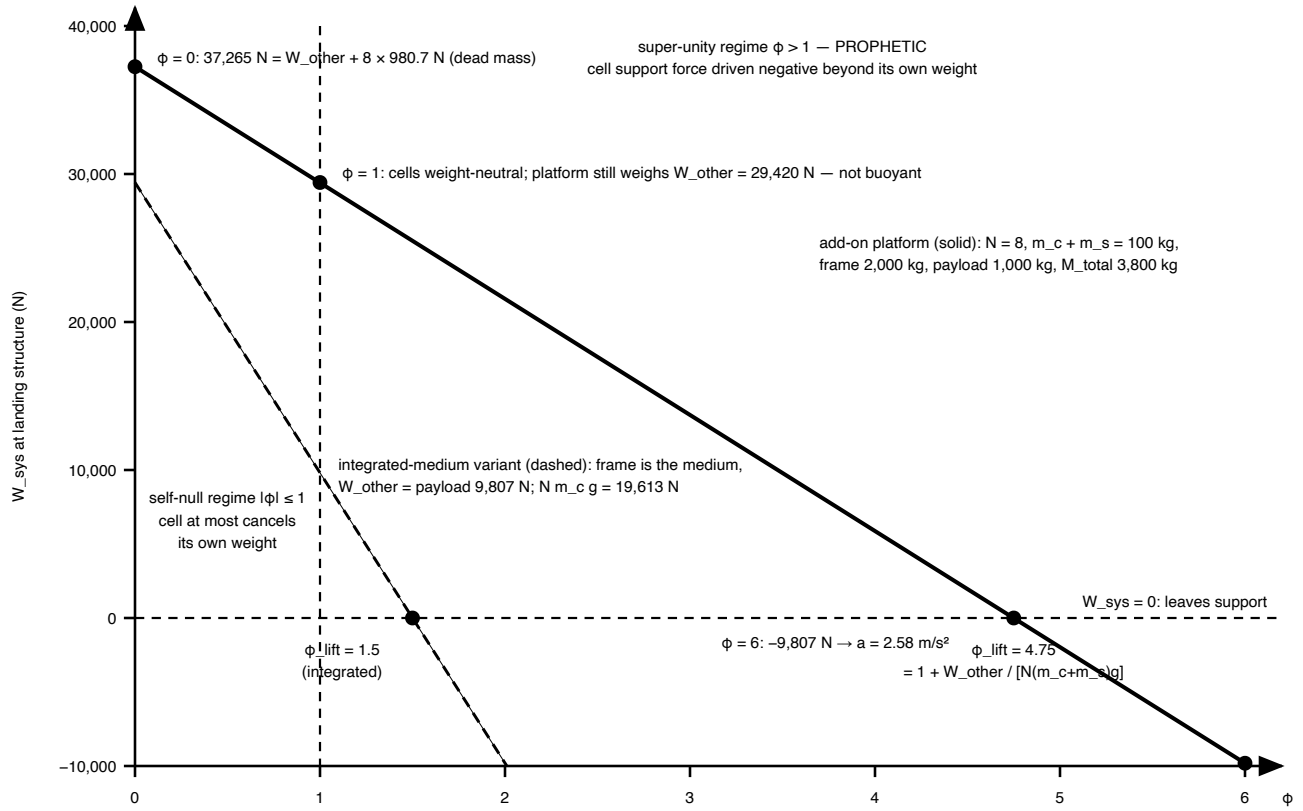


FIG. 11

Three-tier architecture and layered scope — Tier 2 factory qualification (gates Q1–Q8) producing the cell map and array allocation map → qualified cell class or null reference → Tier 1 deployed operation of families A–F with Tier 3 runtime validation, loss-of-drive posture, and recalibration loop; layers L0, L1, L2 with the gates between them.

Lift budget and regimes — net support force at the landing structure vs offset ratio ϕ (P1 platform)



$$W_{\text{sys}} = W_{\text{other}} + \sum_i [(m_c + m_s) g - \Delta W_i] = W_{\text{other}} + N (m_c + m_s) g (1 - \phi) \cdot \text{lift-off: } \sum \Delta W_i - N (m_c + m_s) g \geq W_{\text{other}}$$

add-on benefit of a cell of mass $m = (\phi - 1) m_g$: negative at zero drive, zero at $\phi = 1$, positive only in the super-unity regime

integrated-medium configuration: the medium is mass carried anyway $\rightarrow W_{\text{other}}$ shrinks by its weight; with both signs mapped a cell of mass m supplies $2 m g$ of authority

(from $+m g$ heavier to $-m g$ lighter) — the basis of the ballast (P4), elevator balancing (P2), hook-block, and adapter (P3) embodiments in the self-null regime

inertial mass taken as unchanged: $a = [\sum \Delta W_i - N(m_c + m_s)g - W_{\text{other}}] / M_{\text{total}}$ once unsupported; climb throughput $P \geq M_{\text{total}} g v / \eta$ (η from the allocation map)

example: 3,800 kg at 1 m/s $\rightarrow P \geq 37.3$ kW/ η . Hold power and η are map outputs, not predicted. Neither regime is presumed reachable; ϕ_{lift} is not represented as achieved.

$g = 9.80665$ m/s²; every number follows from the stated masses (P1)

FIG. 12

Lift budget and regimes — net support force at the landing structure versus offset ratio ϕ for the P1 platform (solid: add-on cells; dashed: integrated-medium configuration); self-null and super-unity regimes; lift-off threshold $\phi_{\text{lift}} = 1 + W_{\text{other}} / [N (m_c + m_s) g]$ (4.75 and 1.5 in the two variants).